Engineering Mechanics: Statics

14th Edition
SI Units

Chapter 6: Structural Analysis

Chapter Objectives

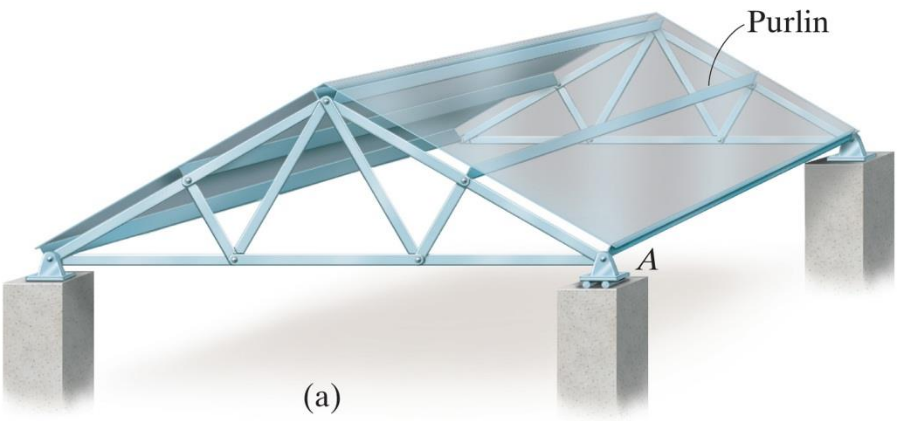
- To show how to determine the forces in the members of a truss using the method of joints and the method of sections
- To analyze the forces acting on the members of frames and machines composed of pin-connected members

Chapter Outline

6.1 Simple Trusses

A truss is a structure composed of slender members joined together at their end points

Planar trusses used to support roofs and bridges

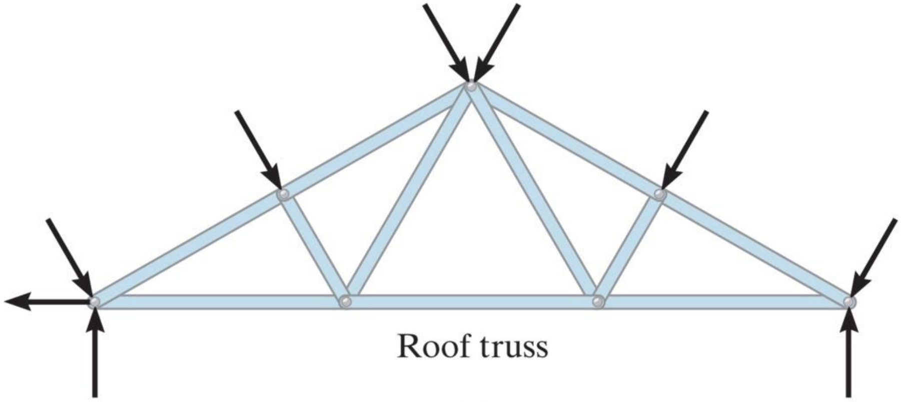


Roof load is transmitted to the truss at the joints by means of a series of purlins

6.1 Simple Trusses

A truss is a structure composed of slender members joined together at their end points

Planar trusses used to support roofs and bridges

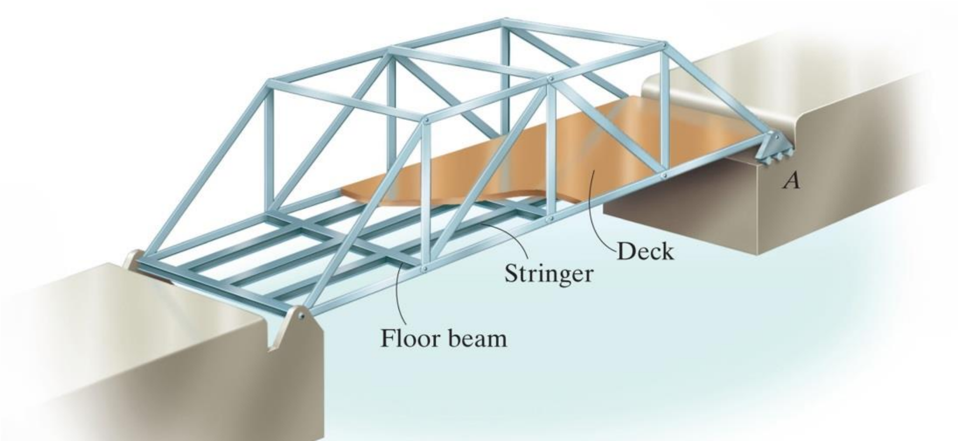


(b)

The analysis of the forces developed in the truss members is 2D

6.1 Simple Trusses

- ✓ For a bridge, the load on the deck is first transmitted to stringers, then to floor beams, and finally to the joints of the two supporting side trusses
- ✓ Like the roof truss, the bridge truss loading is also coplanar



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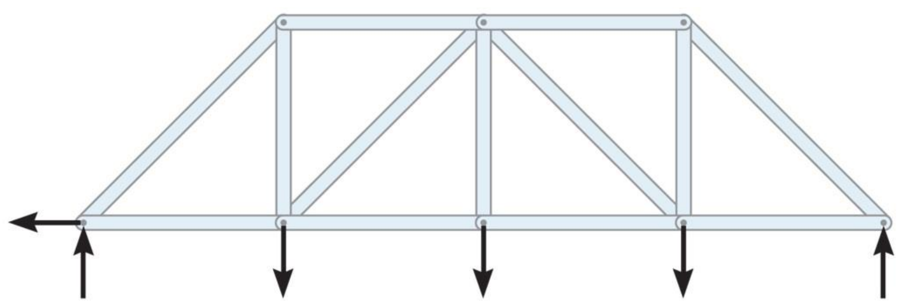
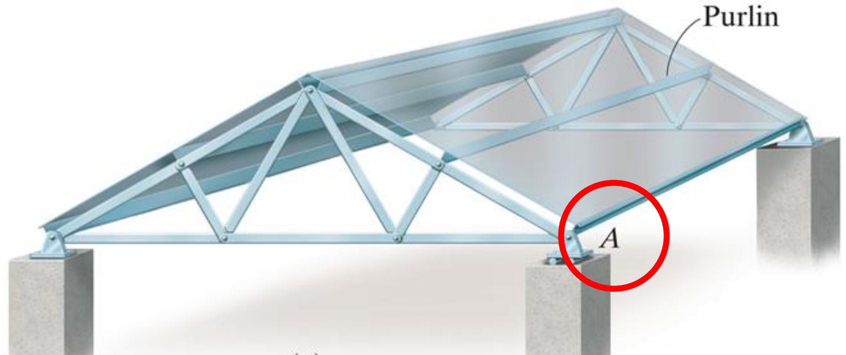
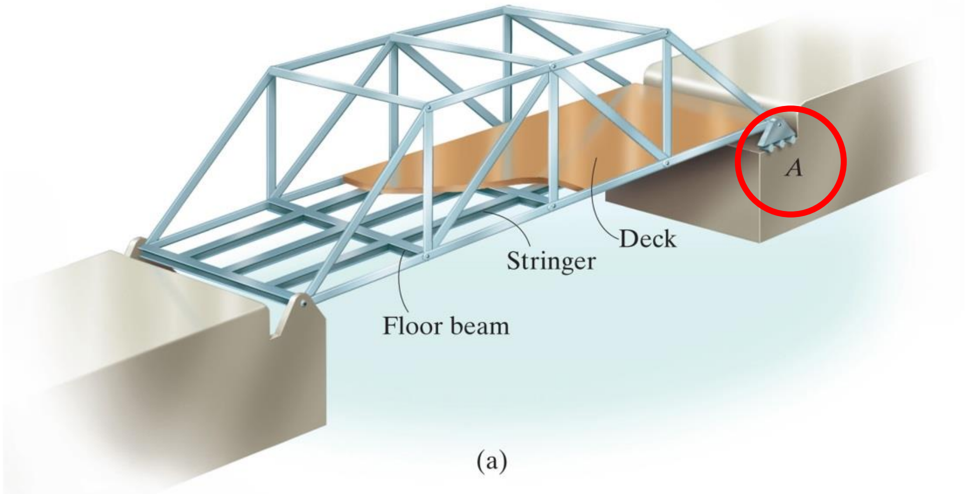

Bridge truss
(b)

6.1 Simple Trusses

- When bridge or roof trusses extend over large distances, a rocker or roller is commonly used for supporting one end, Eg: joint A
- This type of support allows freedom for expansion or contraction of the members due to a change in temperature or application of loads



(a)



(a)

6.1 Simple Trusses

Assumptions for Design

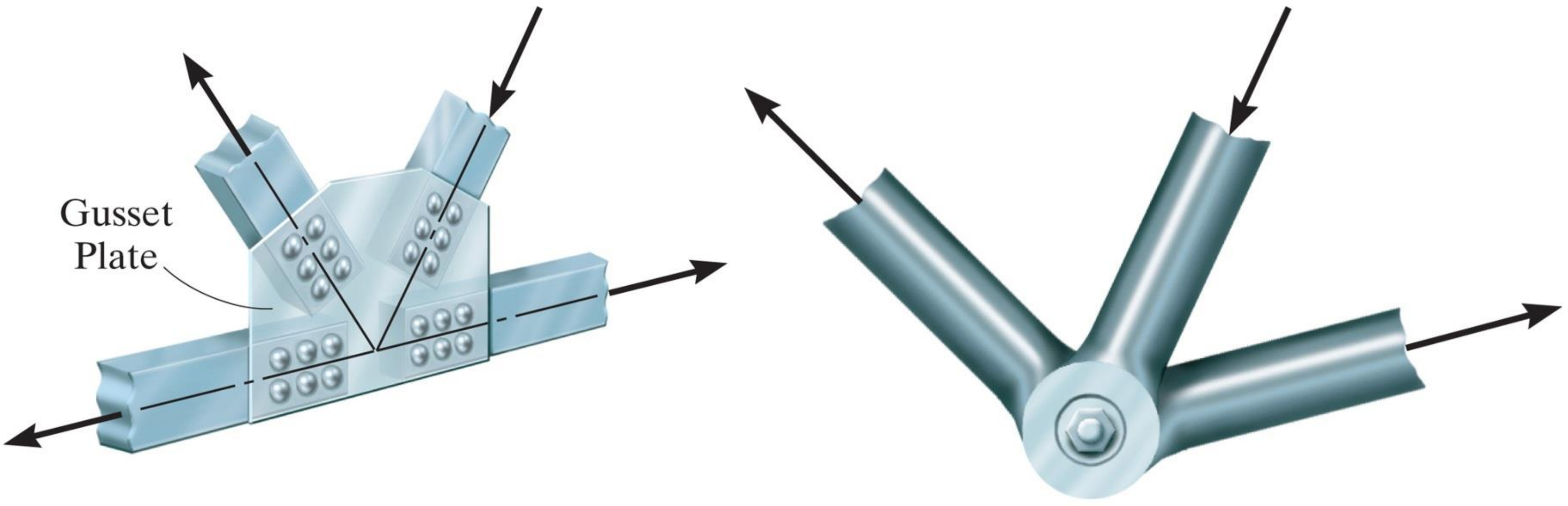
- ***All loadings are applied at the joints***

  - the weight of the members is neglected

- ***The members are joined together by smooth pins***

  - Assume connections provided the center lines of the joining members are *concurrent*

6.1 Simple Trusses



(a)

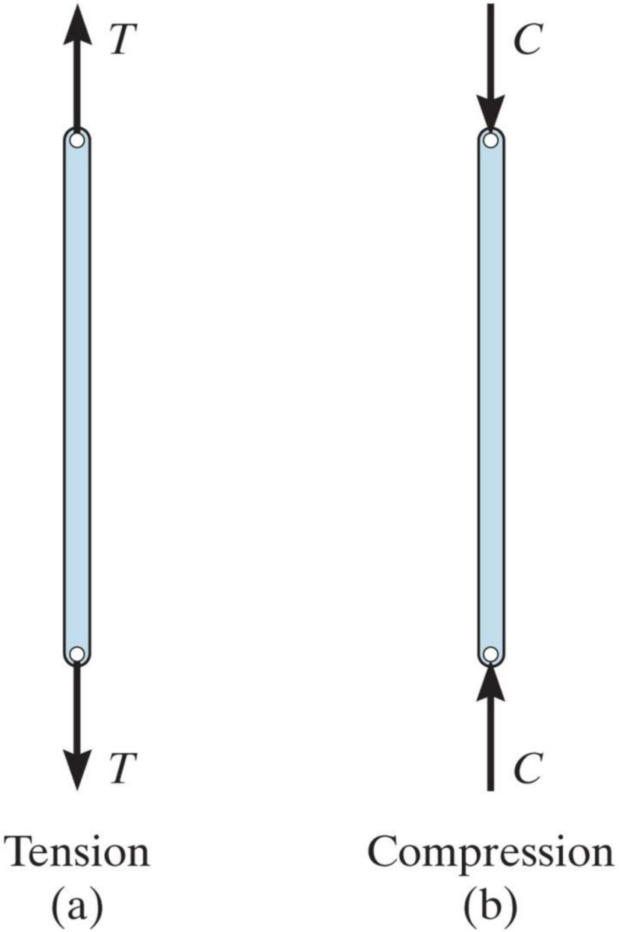
(b)

- ***The members are joined together by smooth pins***
  - Assume connections provided the center lines of the joining members are *concurrent*

6.1 Simple Trusses

Because of these two assumptions, each truss member will act as a two force-member

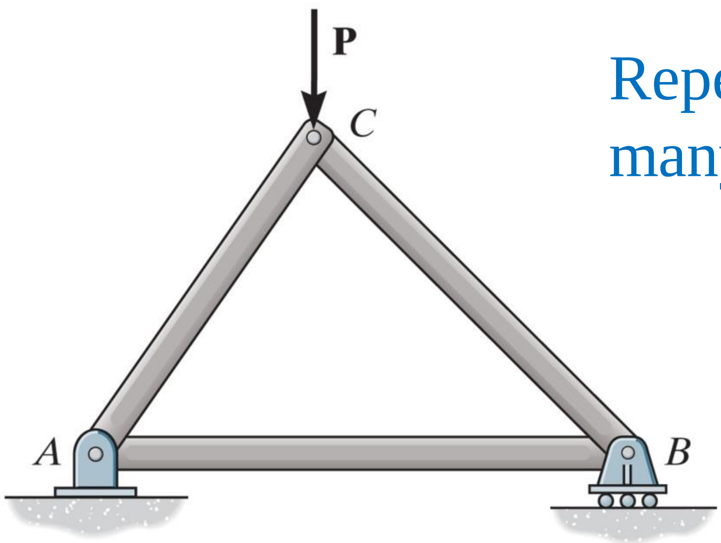
- ✓ If the force tends to elongate the member, it is a *tensile force* (T)
- ✓ if the force tends to shorten the member, it is a *compressive force* (C)



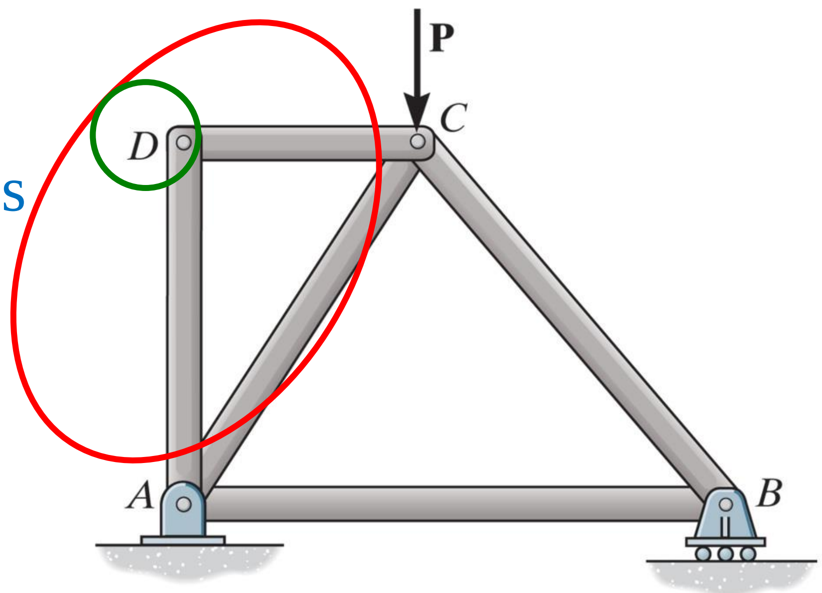
6.1 Simple Trusses

Simple Truss

- ✓ Triangular truss → rigid
- ✓ Simple truss: a truss can be constructed by expanding the basic triangular truss



Repeated
many times



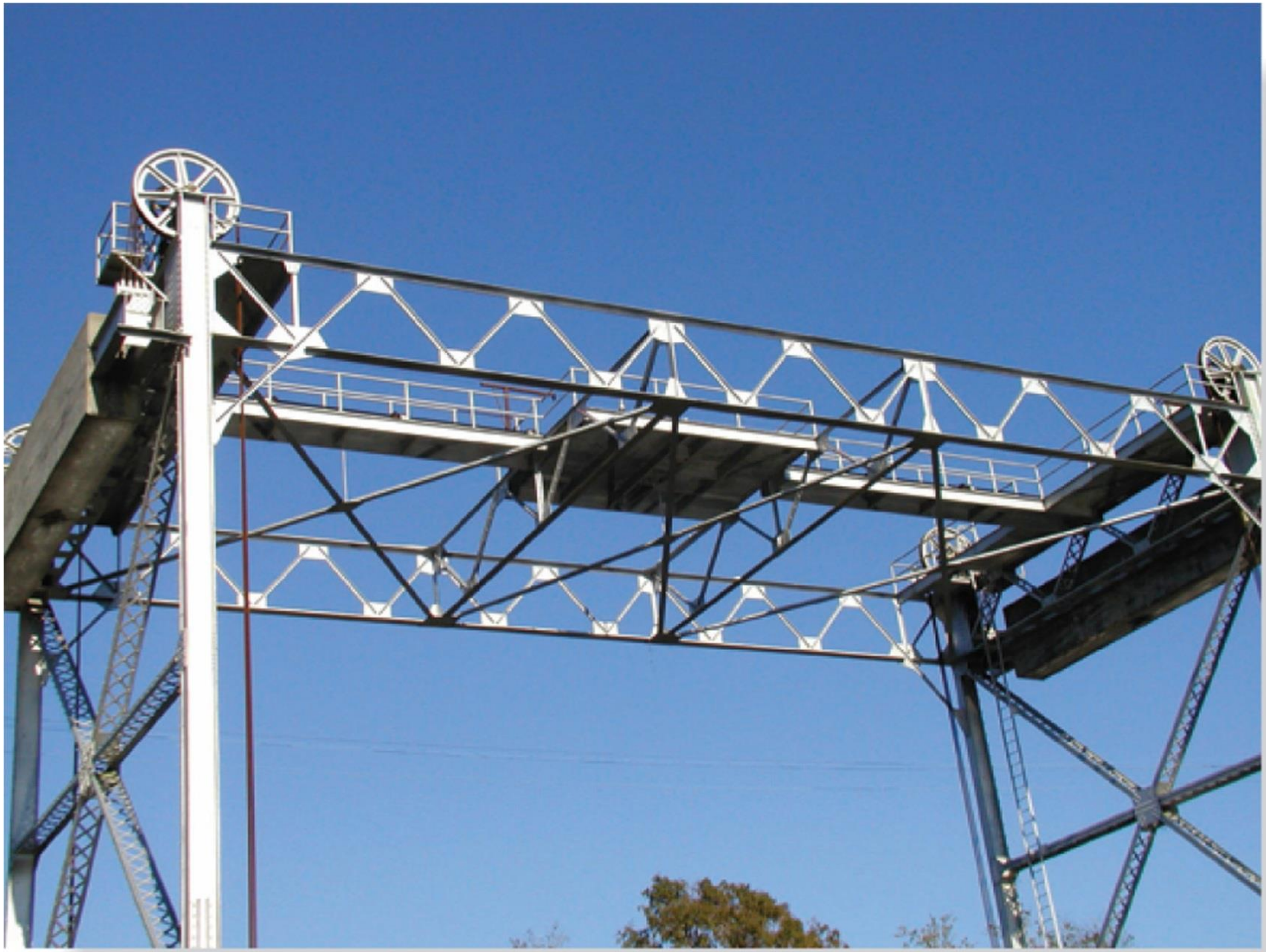
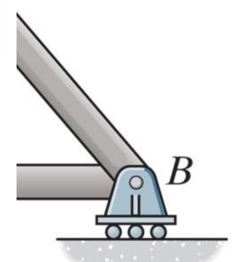


Figure: 06_PH001

The use of metal gusset plates in the construction of these Warren trusses is clearly evident.

ucted

uss



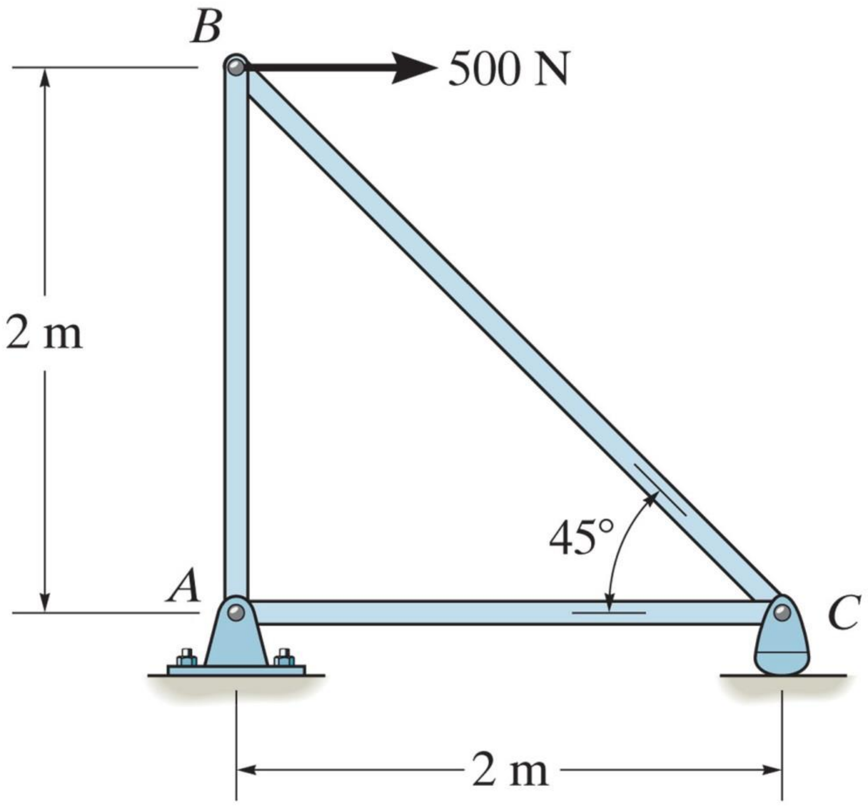
6.2 The Method of Joints

For truss, we need to know the force in each members

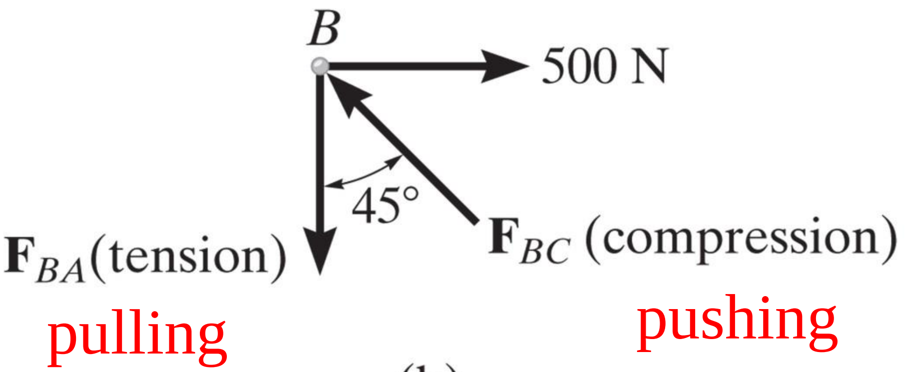
The method of joints

- ✓ The entire truss and each of its joints are in equilibrium → force equilibrium equations can be used to obtain the member forces
- ✓ Force system acting at each joint is coplanar and concurrent
- ✓ $\sum F_x = 0$ and $\sum F_y = 0$

6.2 The Method of Joints



(a)



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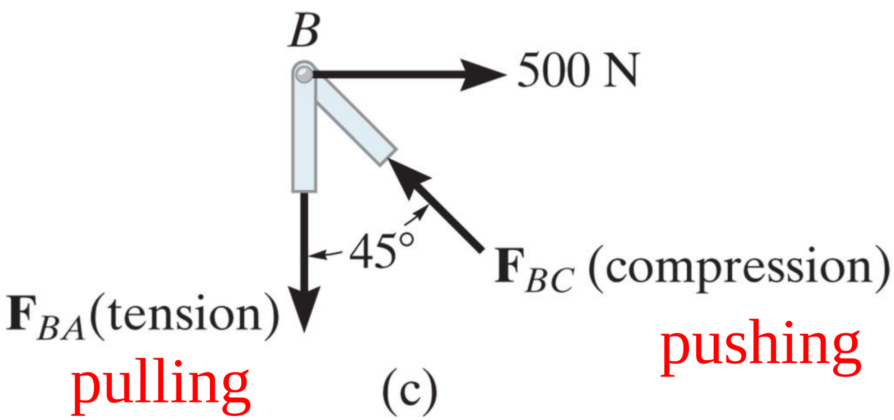


(c)

6.2 The Method of Joints

At least one known fore and at most two unknown forces → using $\sum F_x = 0$ and $\sum F_y = 0$

Two methods to determine the Correct Sense of the unknown member force

1. The correct sense of a direction of an unknown force can be determined "by inspection"

6.2 The Method of Joints

Fig 6-7

- $\mathbf{F}_{BC}$ must push on the pin (compression) since its horizontal component must balance the 500N force
- $\mathbf{F}_{BA}$ is a tensile force since it balances the vertical component of $\mathbf{F}_{BC}$

In more complicated problems, the sense of the member can be assumed

- A positive answer indicates that the sense is correct, whereas a negative answer indicates that the sense must be reversed

6.2 The Method of Joints

2. Always assume the unknown member forces acting on the joint's FBD to be in tension. (pull)

   - Positive scalars→ members in tension

   - Negative scalars→ members in compression

# of Joints

ıember forces
n tension. (pull)

tension

n compression

Figure: 06_PH002
The forces in the members of this simple roof truss can be determined using the method of joints.

Procedure for analysis

The following procedure provides a means for analyzing a truss using the method of joints.

- Draw the free-body diagram of a joint having at least one known force and at most two unknown forces. (If this joint is at one of the supports, then it may be necessary first to calculate the external reactions at the support.)

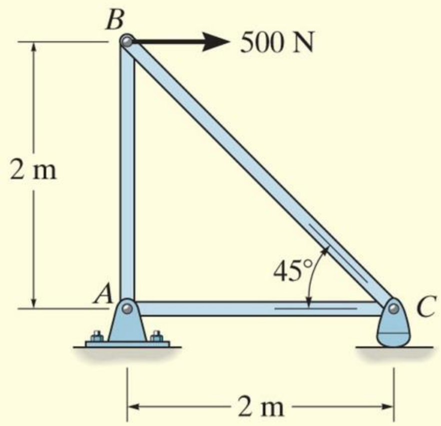
- Use one of the two methods described above for establishing the sense of an unknown force.

Procedure for analysis

- Orient the x and y axes such that the forces on the free-body diagram can be easily resolved into their x and y components and then apply the two force equilibrium equations $\Sigma F_x = 0$ and $\Sigma F_y = 0$. Solve for the two unknown member forces and verify their correct sense.

- Using the calculated results, continue to analyze each of the other joints. Remember that a member in *compression* "pushes" on the joint and a member in *tension* "pulls" on the joint. Also, be sure to choose a joint having at most two unknowns and at least one known force.

EXAMPLE 6.1



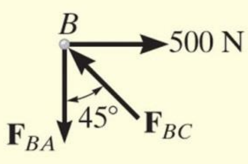
(a)

Determine the force in each member of the truss shown in Fig. 6–8*a* and indicate whether the members are in tension or compression.

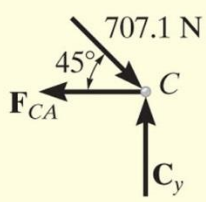
SOLUTION

Since we should have no more than two unknown forces at the joint and at least one known force acting there, we will begin our analysis at joint *B*.

EXAMPLE 6.1 CONTINUED



(b)



(c)

Joint B. The free-body diagram of the joint at B is shown in Fig. 6–8b. Applying the equations of equilibrium, we have

$$\xrightarrow{+} \Sigma F_x = 0; \quad 500 \text{ N} - F_{BC} \sin 45° = 0 \quad F_{BC} = 707.1 \text{ N (C)} \quad \textit{Ans.}$$

$$+\uparrow \Sigma F_y = 0; \quad F_{BC} \cos 45° - F_{BA} = 0 \quad F_{BA} = 500 \text{ N (T)} \quad \textit{Ans.}$$

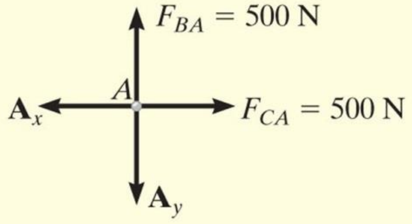
Since the force in member BC has been calculated, we can proceed to analyze joint C to determine the force in member CA and the support reaction at the rocker.

Joint C. From the free-body diagram of joint C, Fig. 6–8c, we have

$$\xrightarrow{+} \Sigma F_x = 0; \quad -F_{CA} + 707.1 \cos 45° \text{ N} = 0 \quad F_{CA} = 500 \text{ N (T)} \quad \textit{Ans.}$$

$$+\uparrow \Sigma F_y = 0; \quad C_y - 707.1 \sin 45° \text{ N} = 0 \quad C_y = 500 \text{ N} \quad \textit{Ans.}$$

EXAMPLE 6.1 CONTINUED



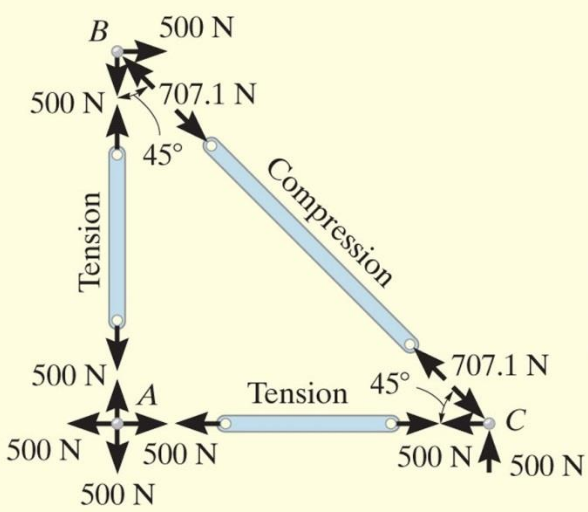
(d)

Joint A. Although it is not necessary, we can determine the components of the support reactions at joint A using the results of F_{CA} and F_{BA}. From the free-body diagram, Fig. 6–8*d*, we have

$$\xrightarrow{+} \Sigma F_x = 0; \qquad 500 \text{ N} - A_x = 0 \qquad A_x = 500 \text{ N}$$

$$+\uparrow \Sigma F_y = 0; \qquad 500 \text{ N} - A_y = 0 \qquad A_y = 500 \text{ N}$$

NOTE: The results of the analysis are summarized in Fig. 6–8*e*. Note that the free-body diagram of each joint (or pin) shows the effects of all the connected members and external forces applied to the joint, whereas the free-body diagram of each member shows only the effects of the end joints on the member.



(e)

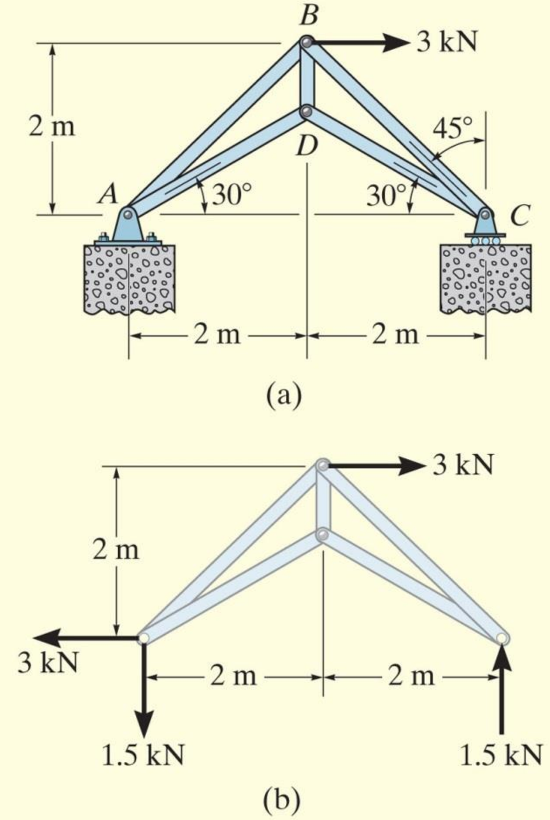
Fig. 6–8

EXAMPLE 6.2

Determine the forces acting in all the members of the truss shown in Fig. 6–9*a*.

SOLUTION

By inspection, there are more than two unknowns at each joint. Consequently, the support reactions on the truss must first be determined. Show that they have been correctly calculated on the free-body diagram in Fig. 6–9*b*. We can now begin the analysis at joint *C*. Why?

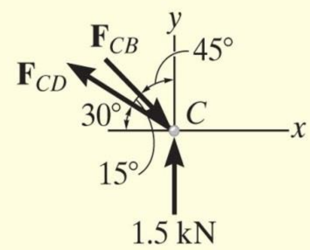


EXAMPLE 6.2 CONTINUED

Joint C. From the free-body diagram, Fig. 6–9*c*,

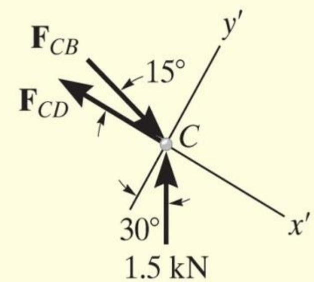
$$\xrightarrow{+} \Sigma F_x = 0; \qquad -F_{CD}\cos 30^\circ + F_{CB}\sin 45^\circ = 0$$

$$+\uparrow \Sigma F_y = 0; \qquad 1.5\text{ kN} + F_{CD}\sin 30^\circ - F_{CB}\cos 45^\circ = 0$$



(c)

These two equations must be solved *simultaneously* for each of the two unknowns. Note, however, that a *direct solution* for one of the unknown forces may be obtained by applying a force summation along an axis that is *perpendicular* to the direction of the other unknown force. For example, summing forces along the y' axis, which is perpendicular to the direction of $\mathbf{F}_{CD}$, Fig. 6–9*d*, yields a *direct solution* for F_{CB}.



(d)

$$+\nearrow \Sigma F_{y'} = 0; \qquad 1.5\cos 30^\circ\text{ kN} - F_{CB}\sin 15^\circ = 0$$

$$F_{CB} = 5.019\text{ kN} = 5.02\text{ kN (C)} \qquad \textit{Ans.}$$

EXAMPLE 6.2 CONTINUED

Then,

$$+\searrow \Sigma F_{x'} = 0;$$

$$-F_{CD} + 5.019 \cos 15^\circ - 1.5 \sin 30^\circ = 0; \quad F_{CD} = 4.10 \text{ kN (T)} \quad \textit{Ans.}$$

Joint *D*. We can now proceed to analyze joint D. The free-body diagram is shown in Fig. 6–9e.

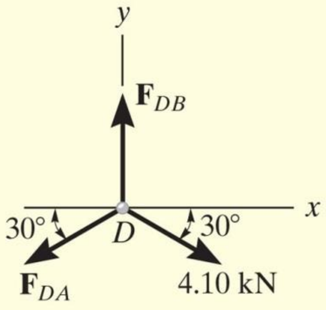
$$\overset{+}{\rightarrow} \Sigma F_x = 0; \qquad -F_{DA} \cos 30^\circ + 4.10 \cos 30^\circ \text{ kN} = 0$$

$$F_{DA} = 4.10 \text{ kN} \quad \text{(T)} \qquad \textit{Ans.}$$

$$+\uparrow \Sigma F_y = 0; \qquad F_{DB} - 2(4.10 \sin 30^\circ \text{ kN}) = 0$$

$$F_{DB} = 4.10 \text{ kN} \quad \text{(T)} \qquad \textit{Ans.}$$

NOTE: The force in the last member, BA, can be obtained from joint B or joint A. As an exercise, draw the free-body diagram of joint B, sum the forces in the horizontal direction, and show that $F_{BA} = 0.776$ kN (C).



(e)

Fig. 6–9

EXAMPLE 6.3

Determine the force in each member of the truss shown in Fig. 6–10a. Indicate whether the members are in tension or compression.

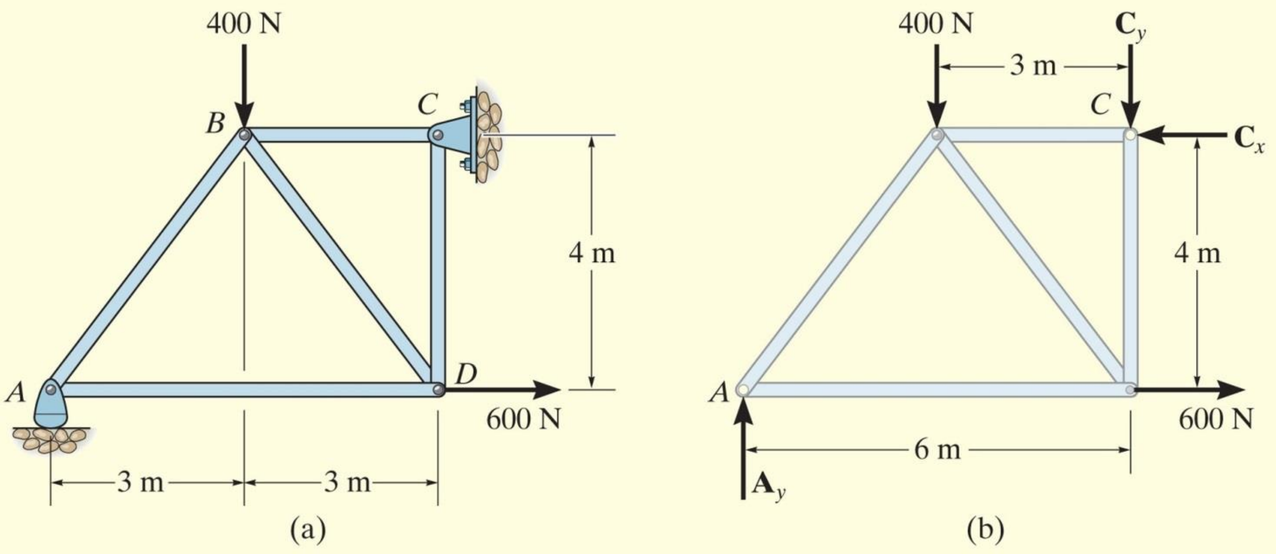


Fig. 6–10

EXAMPLE 6.3 CONTINUED

SOLUTION

Support Reactions. No joint can be analyzed until the support reactions are determined, because each joint has at least three unknown forces acting on it. A free-body diagram of the entire truss is given in Fig. 6–10*b*. Applying the equations of equilibrium, we have

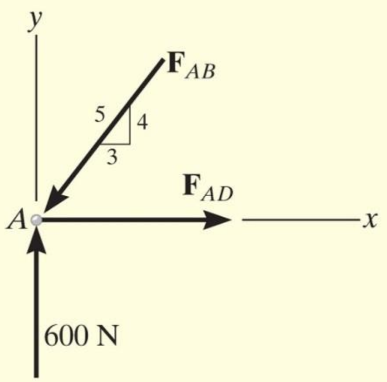
$$\xrightarrow{+} \Sigma F_x = 0; \qquad 600\text{ N} - C_x = 0 \qquad C_x = 600\text{ N}$$

$$\circlearrowleft + \Sigma M_C = 0; \qquad -A_y(6\text{ m}) + 400\text{ N}(3\text{ m}) + 600\text{ N}(4\text{ m}) = 0$$

$$A_y = 600\text{ N}$$

$$+\uparrow \Sigma F_y = 0; \qquad 600\text{ N} - 400\text{ N} - C_y = 0 \qquad C_y = 200\text{ N}$$

The analysis can now start at either joint *A* or *C*. The choice is arbitrary since there are one known and two unknown member forces acting on the pin at each of these joints.



(c)

Joint A. (Fig. 6–10*c*). As shown on the free-body diagram, $\mathbf{F}_{AB}$ is assumed to be compressive and $\mathbf{F}_{AD}$ is tensile. Applying the equations of equilibrium, we have

$$+\uparrow \Sigma F_y = 0; \qquad 600\text{ N} - \tfrac{4}{5}F_{AB} = 0 \qquad F_{AB} = 750\text{ N (C)} \qquad \textit{Ans.}$$

$$\xrightarrow{+} \Sigma F_x = 0; \qquad F_{AD} - \tfrac{3}{5}(750\text{ N}) = 0 \qquad F_{AD} = 450\text{ N (T)} \qquad \textit{Ans.}$$

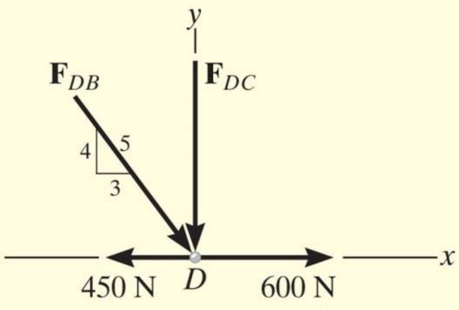
EXAMPLE 6.3 CONTINUED

Joint D. (Fig. 6–10*d*). Using the result for F_{AD} and summing forces in the horizontal direction, Fig. 6–10*d*, we have

$$\overset{+}{\rightarrow} \Sigma F_x = 0; \qquad -450\text{ N} + \tfrac{3}{5}F_{DB} + 600\text{ N} = 0 \qquad F_{DB} = -250\text{ N}$$

The negative sign indicates that $\mathbf{F}_{DB}$ acts in the *opposite sense* to that shown in Fig. 6–10*d*.* Hence,

$$F_{DB} = 250\text{ N (T)} \qquad \textit{Ans.}$$



(d)

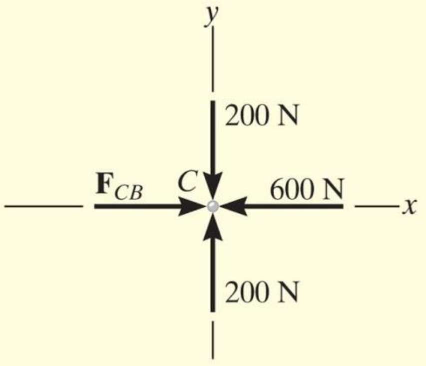
To determine $\mathbf{F}_{DC}$, we can either correct the sense of $\mathbf{F}_{DB}$ on the free-body diagram, and then apply $\Sigma F_y = 0$, or apply this equation and retain the negative sign for F_{DB}, i.e.,

$$+\uparrow \Sigma F_y = 0; \qquad -F_{DC} - \tfrac{4}{5}(-250\text{ N}) = 0 \qquad F_{DC} = 200\text{ N} \quad \text{(C)} \qquad \textit{Ans.}$$

Joint C. (Fig. 6–10*e*).

$$\overset{+}{\rightarrow} \Sigma F_x = 0; \qquad F_{CB} - 600\text{ N} = 0 \qquad F_{CB} = 600\text{ N} \quad \text{(C)} \qquad \textit{Ans.}$$

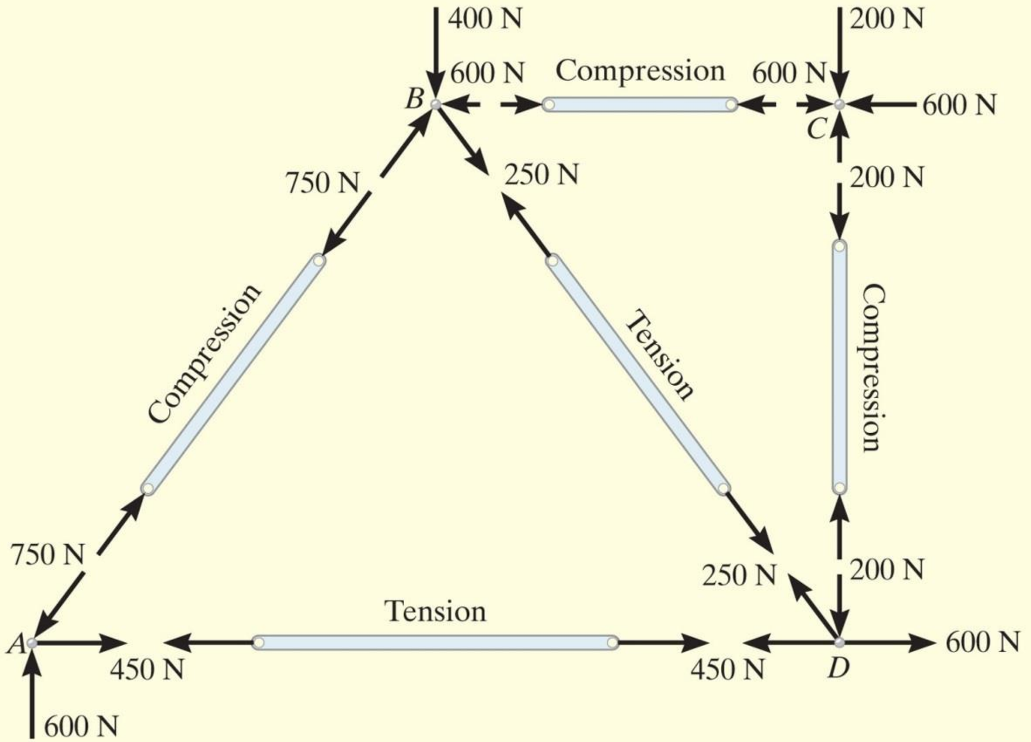
$$+\uparrow \Sigma F_y = 0; \qquad 200\text{ N} - 200\text{ N} \equiv 0 \quad \text{(check)}$$



(e)

EXAMPLE 6.3 CONTINUED

NOTE: The analysis is summarized in Fig. 6–10*f*, which shows the free-body diagram for each joint and member.



(f)

Fig. 6–10 (cont.)

*The proper sense could have been determined by inspection, prior to applying $\Sigma F_x = 0$.

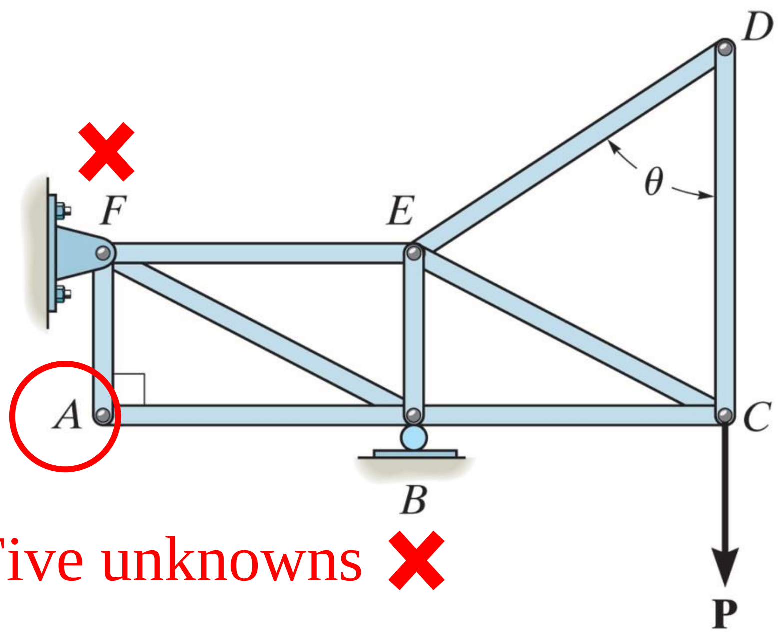
6.3 Zero-Force Members

- ✓ Truss analysis using the method of joints is greatly simplified if we can first identify those members which support *no loading*.

- ✓ These zero-force members are used to increase the stability of the truss during construction and to provide added support if the loading is changed.

6.3 Zero-Force Members

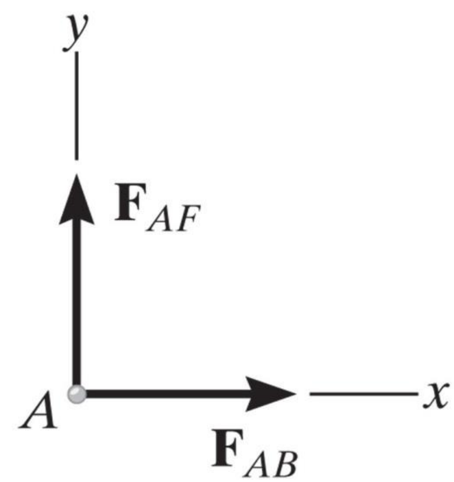
- ✓ The zero-force members →found by inspection of each of the joints.



Five unknowns ✗

(a)

FBD

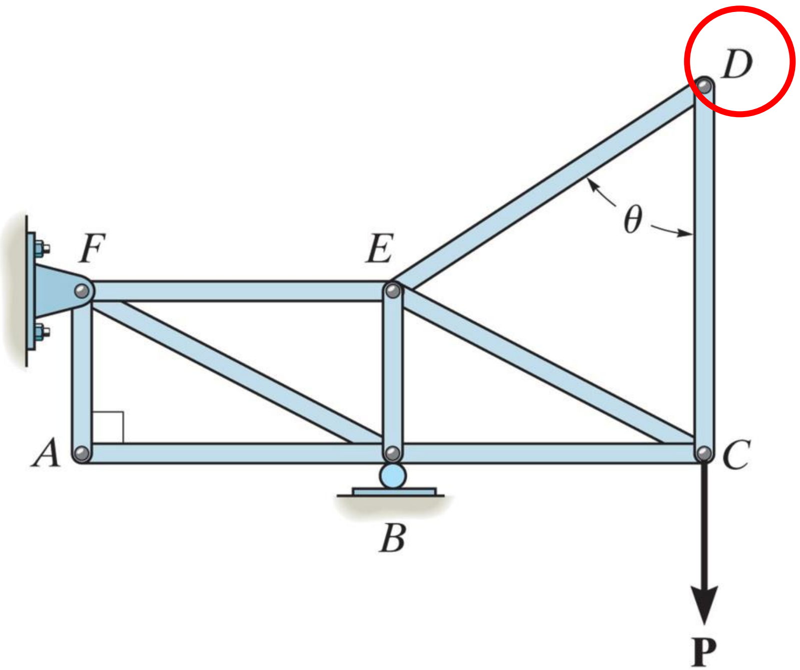


$$\xrightarrow{+} \Sigma F_x = 0; \quad F_{AB} = 0$$

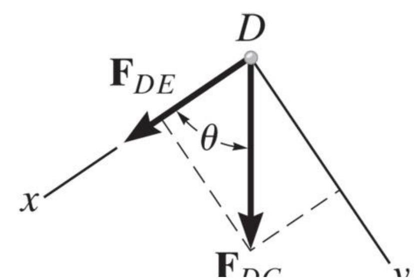
$$+\uparrow \Sigma F_y = 0; \quad F_{AF} = 0$$

Members AB and AF are zero-force members.

6.3 Zero-Force Members



(a)



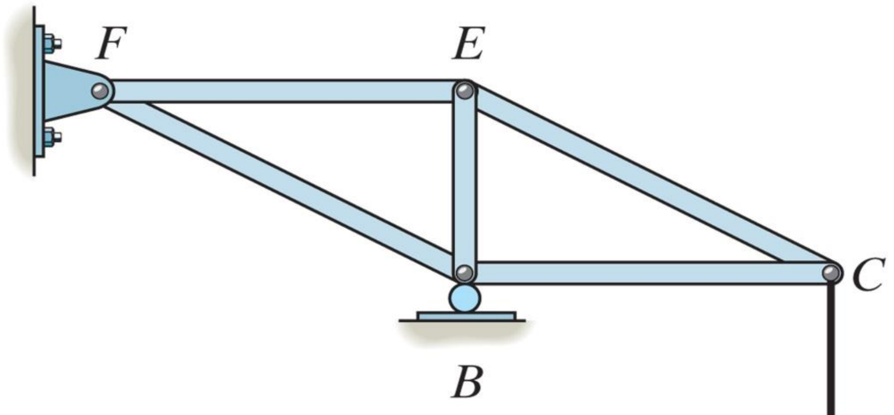
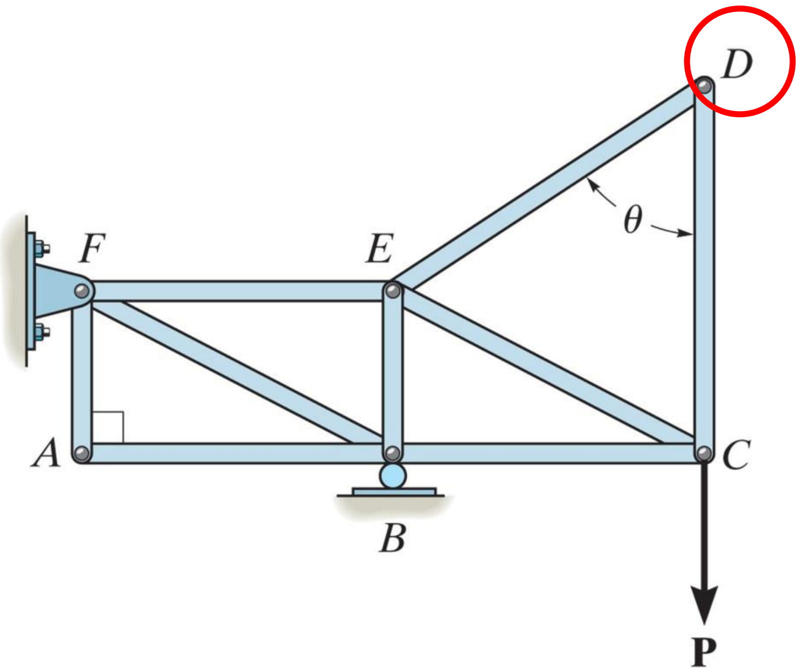
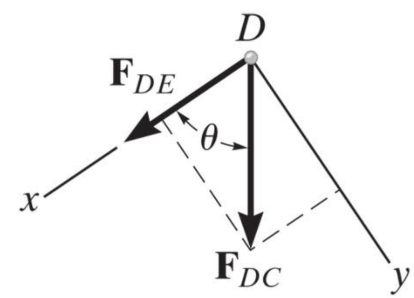


Fig (a) supported by only five members

6.3 Zero-Force Members



(a)



$$+\searrow \Sigma F_y = 0;\ F_{DC}\sin\theta = 0;\quad F_{DC} = 0 \text{ since } \sin\theta \neq 0$$

$$+\swarrow \Sigma F_x = 0;\ F_{DE} + 0 = 0;\quad F_{DE} = 0$$

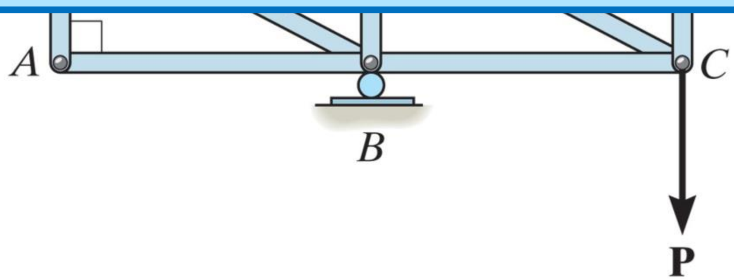
(c)

Members DC and DE are zero-force members.

6.3 Zero-Force Members

Conclusion

If only two non-collinear members form a truss joint and no external load or support reactions is applied to the joint, the two members must be zero-force members.



(a)

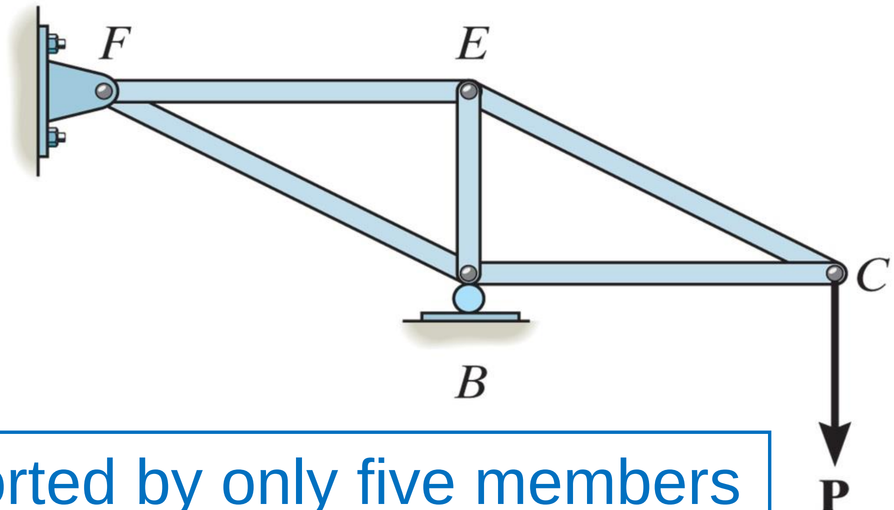


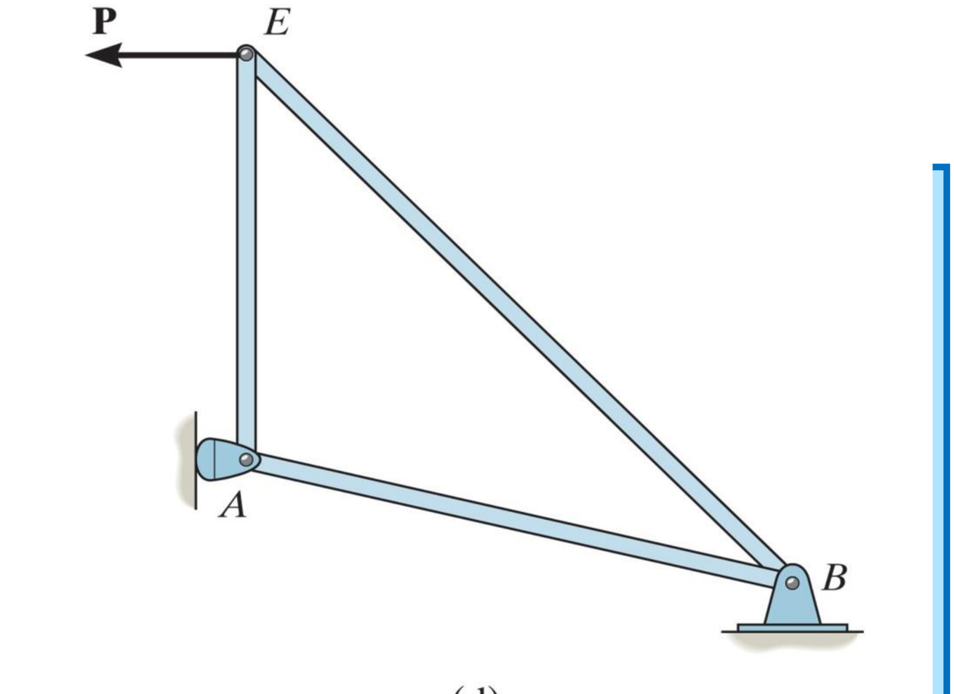
Fig (a) supported by only five members

6.3 Zero-Force Members

- ✓ From the FBD of the pin of the joint D, DA is a zero-force member

supporting the load **P**

- ✓ From the FBD of the pi
zero-force member



(d)

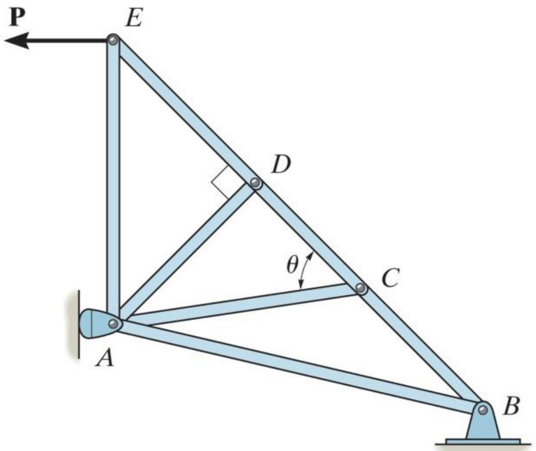
In General

if three members form a trus
members are collinear, the t
member provided no externa
is applied to the joint

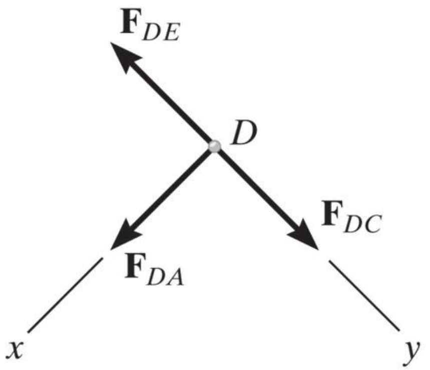
6.3 Zero-Force Members

- ✓ From the FBD of the pin of the joint D, DA is a zero-force member

- ✓ From the FBD of the pin of the joint C, CA is a zero-force member

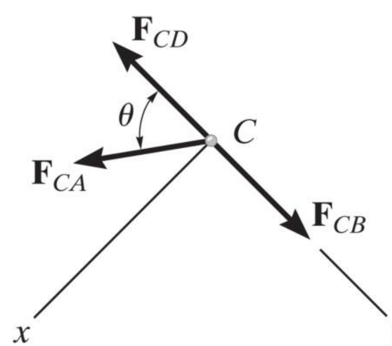


(a)



$$+\swarrow \Sigma F_x = 0; \quad F_{DA} = 0$$
$$+\searrow \Sigma F_y = 0; \quad F_{DC} = F_{DE}$$

(b)



$$+\swarrow \Sigma F_x = 0; \quad F_{CA} \sin\theta = 0; \quad F_{CA} = 0 \text{ since } \sin\theta \neq 0;$$
$$+\searrow \Sigma F_y = 0; \quad F_{CB} = F_{CD}$$

(c)

6.3 Zero-Force Members

- ✓ From the FBD of the pin of the joint D, DA is a zero-force member

- ✓ From the FBD of the pin of the joint C, CA is a zero-force member

In General

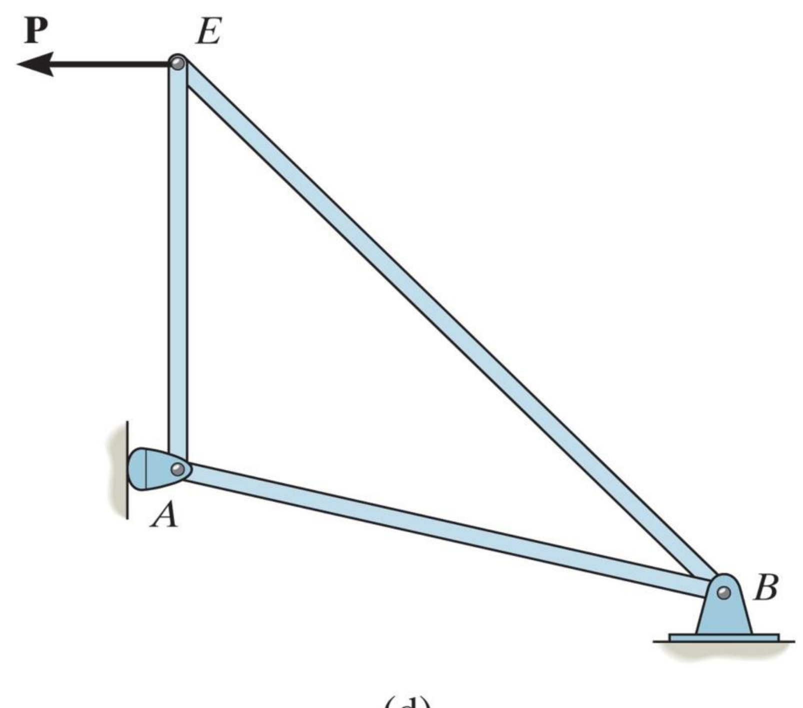
if three members form a truss joint for which two of the members are collinear, the third member is a zero-force member provided no external force or support reaction is applied to the joint

6.3 Zero-Force Members

- ✓ From the FBD of the pin of the joint D, DA is a zero-force member

- ✓ From the FBD of the pi zero-force member

supporting the load **P**



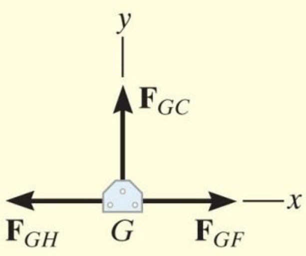
(d)

In General

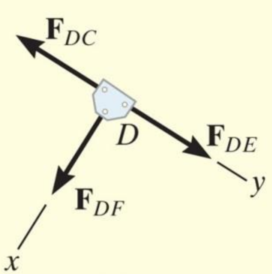
if three members form a trus members are collinear, the t member provided no externa is applied to the joint

EXAMPLE 6.4

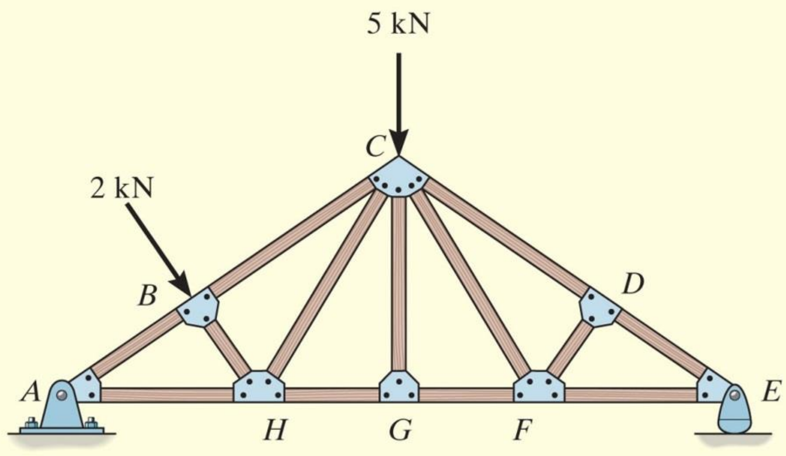
Using the method of joints, determine all the zero-force members of the *Fink roof truss* shown in Fig. 6–13*a*. Assume all joints are pin connected.



(b)



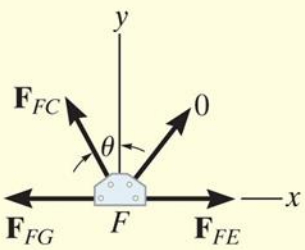
(c)



(a)

Fig. 6–13

EXAMPLE 6.4 CONTINUED



(d)

SOLUTION

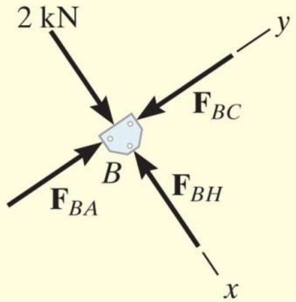
Look for joint geometries that have three members for which two are collinear. We have

Joint G. (Fig. 6–13*b*).

$$+\uparrow \Sigma F_y = 0; \qquad F_{GC} = 0 \qquad \textit{Ans.}$$

Realize that we could not conclude that GC is a zero-force member by considering joint C, where there are five unknowns. The fact that GC is a zero-force member means that the 5-kN load at C must be supported by members CB, CH, CF, and CD.

EXAMPLE 6.4 CONTINUED



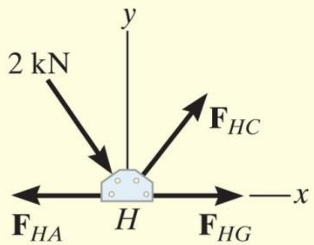
(e)

Joint *D*. (Fig. 6–13*c*).

$$+\swarrow \Sigma F_x = 0; \qquad F_{DF} = 0 \qquad \textit{Ans.}$$

Joint *F*. (Fig. 6–13*d*).

$$+\uparrow \Sigma F_y = 0; \qquad F_{FC}\cos\theta = 0 \quad \text{Since } \theta \neq 90^\circ, \quad F_{FC} = 0 \qquad \textit{Ans.}$$



(f)

NOTE: If joint *B* is analyzed, Fig. 6–13*e*,

$$+\searrow \Sigma F_x = 0; \qquad 2\text{ kN} - F_{BH} = 0 \quad F_{BH} = 2\text{ kN} \quad \text{(C)}$$

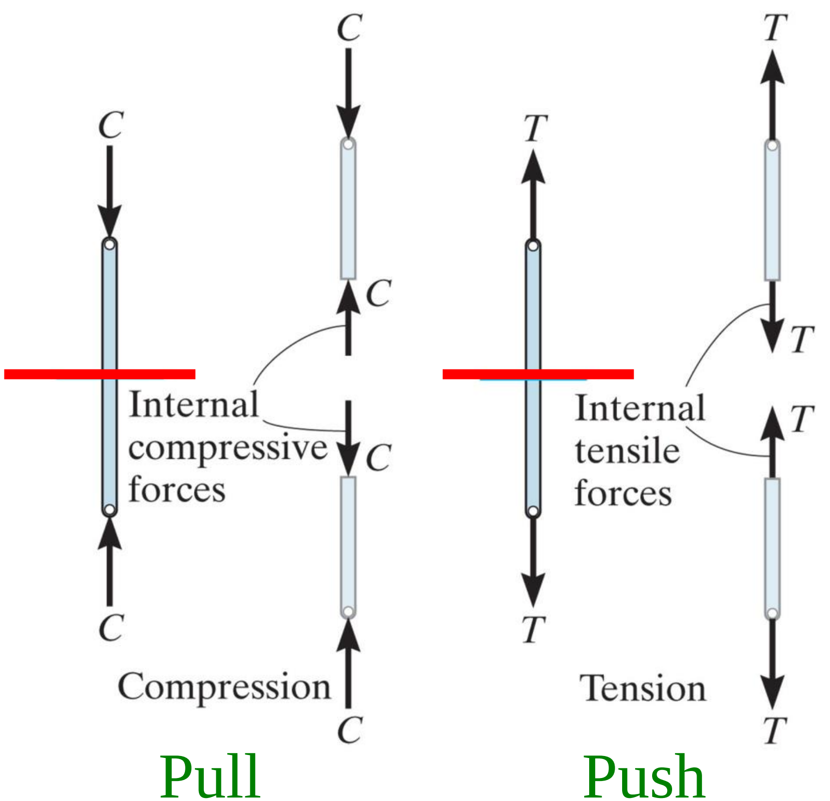
Also, F_{HC} must satisfy $\Sigma F_y = 0$, Fig. 6–13*f*, and therefore *HC* is *not* a zero-force member.

6.4 The Method of Sections

Method of sections

- ✓ If the truss is in equilibrium → any segment of the truss is also in equilibrium

- ✓ To find forces within members, an imaginary section is used to cut each member into 2 parts and "expose" each internal force as "external" to FBD



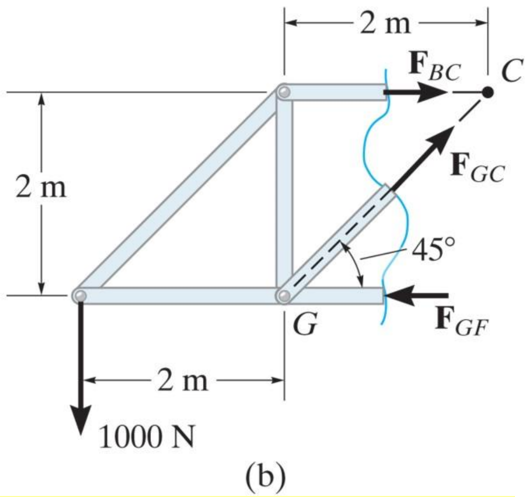
6.4 The Method of Sections

The method of section also be used to “cut” or section the members of an entire truss.

Only three independent equilibrium equations ($\sum F_x = 0$; $\sum F_y = 0$; $\sum M_o = 0$) → select a section no more than three members in which the force are unknown

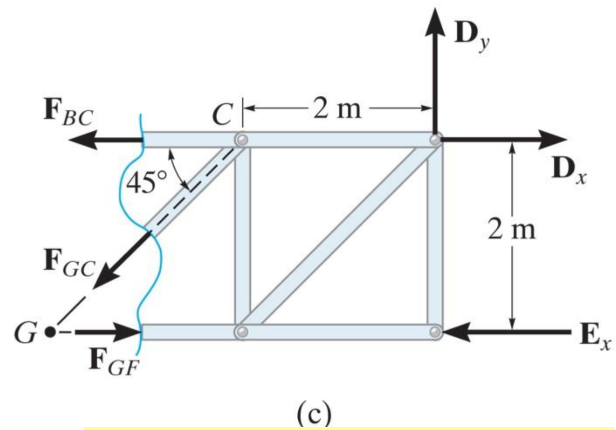
Note:

1. The force in a member is along its axis
2. Member forces are equal and opposite to those acting on the other part – Newton's Law



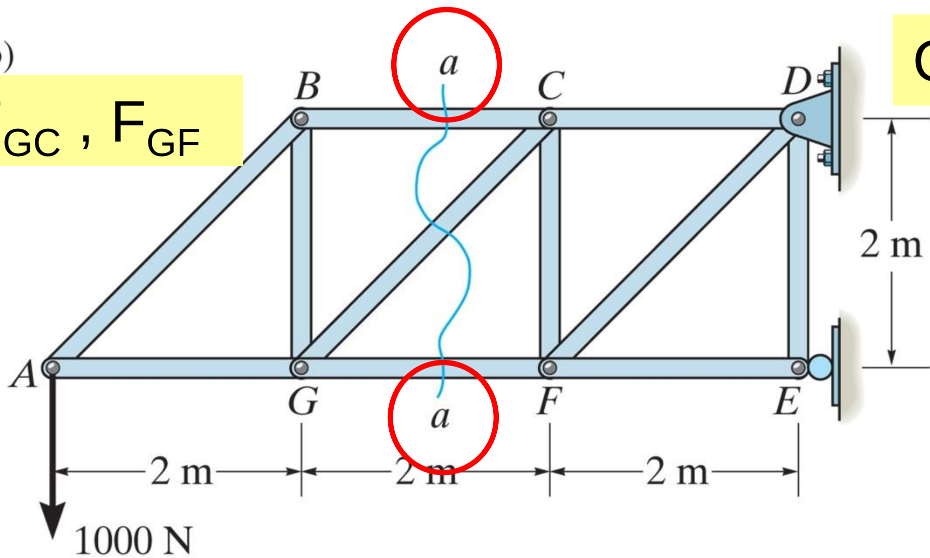
(b)

Obtain: F_{BC} , F_{GC} , F_{GF}



(c)

Obtain: D_x , D_y , E_x



6.4 The Method of Sections

Two ways to determine the correct sense of an unknown member force:

1. "by inspection.“

2. Always assume that the unknown member forces at the cut section are *tensile forces*, i.e., "pulling" on the member. By doing this, the numerical solution of the equilibrium equations will yield *positive scalars for members in tension and negative scalars for members in compression.*

Figure: 06_PH003

The forces in selected members of this Pratt truss can readily be determined using the method of sections.

Figure

The forces in selected members of this Pr
method

Figure: 06_PH004

Simple trusses are often used in the construction of large cranes in order to reduce the weight of the boom and tower.

PEARSON

Procedure for analysis

The forces in the members of a truss may be determined by the method of sections using the following procedure.

Free-Body Diagram.

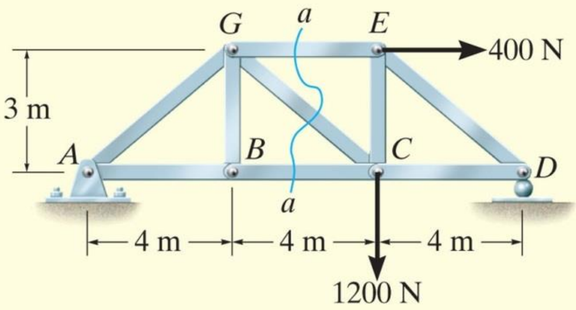
- Make a decision on how to “cut” or section the truss through the members where forces are to be determined.
- Before isolating the appropriate section, it may first be necessary to determine the truss’s support reactions. If this is done then the three equilibrium equations will be available to solve for member forces at the section.
- Draw the free-body diagram of that segment of the sectioned truss which has the least number of forces acting on it.
- Use one of the two methods described above for establishing the sense of the unknown member forces.

Procedure for analysis

Equations of Equilibrium.

- Moments should be summed about a point that lies at the intersection of the lines of action of two unknown forces, so that the third unknown force can be determined directly from the moment equation.
- If two of the unknown forces are *parallel*, forces may be summed *perpendicular* to the direction of these unknowns to determine *directly* the third unknown force.

EXAMPLE 6.5



(a)

Determine the force in members GE, GC, and BC of the truss shown in Fig. 6–16a. Indicate whether the members are in tension or compression.

SOLUTION

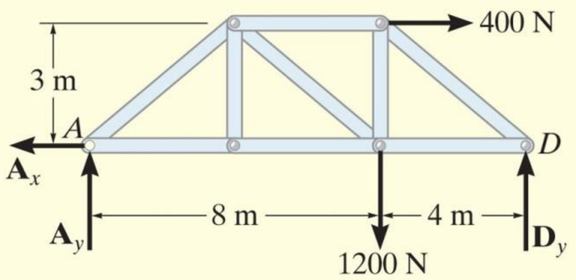
Section aa in Fig. 6–16a has been chosen since it cuts through the *three* members whose forces are to be determined. In order to use the method of sections, however, it is *first* necessary to determine the external reactions at A or D. Why? A free-body diagram of the entire truss is shown in Fig. 6–16b. Applying the equations of equilibrium, we have

$$\overset{+}{\rightarrow}\Sigma F_x = 0; \qquad 400\text{ N} - A_x = 0 \qquad A_x = 400\text{ N}$$

$$\circlearrowleft + \Sigma M_A = 0; \qquad -1200\text{ N}(8\text{ m}) - 400\text{ N}(3\text{ m}) + D_y(12\text{ m}) = 0$$

$$D_y = 900\text{ N}$$

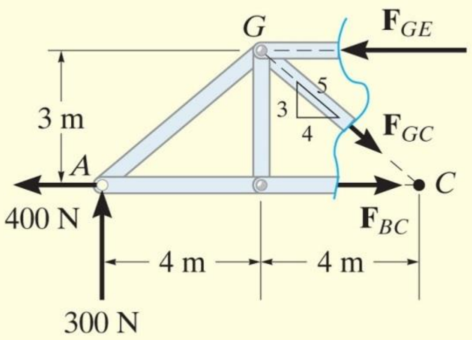
$$+\uparrow \Sigma F_y = 0; \qquad A_y - 1200\text{ N} + 900\text{ N} = 0 \qquad A_y = 300\text{ N}$$



(b)

Free-Body Diagram. For the analysis the free-body diagram of the left portion of the sectioned truss will be used, since it involves the least number of forces, Fig. 6–16c.

EXAMPLE 6.5 CONTINUED



(c)

Fig. 6–16

Equations of Equilibrium. Summing moments about point G eliminates $\mathbf{F}_{GE}$ and $\mathbf{F}_{GC}$ and yields a direct solution for F_{BC}.

$$\circlearrowleft +\Sigma M_G = 0; \quad -300\text{ N}(4\text{ m}) - 400\text{ N}(3\text{ m}) + F_{BC}(3\text{ m}) = 0$$

$$F_{BC} = 800\text{ N} \quad \text{(T)} \qquad \textit{Ans.}$$

In the same manner, by summing moments about point C we obtain a direct solution for F_{GE}.

$$\circlearrowleft +\Sigma M_C = 0; \quad -300\text{ N}(8\text{ m}) + F_{GE}(3\text{ m}) = 0$$

$$F_{GE} = 800\text{ N} \quad \text{(C)} \qquad \textit{Ans.}$$

Since $\mathbf{F}_{BC}$ and $\mathbf{F}_{GE}$ have no vertical components, summing forces in the y direction directly yields F_{GC}, i.e.,

$$+\uparrow \Sigma F_y = 0; \qquad 300\text{ N} - \tfrac{3}{5}F_{GC} = 0$$

$$F_{GC} = 500\text{ N} \quad \text{(T)} \qquad \textit{Ans.}$$

NOTE: Here it is possible to tell, by inspection, the proper direction for each unknown member force. For example, $\Sigma M_C = 0$ requires $\mathbf{F}_{GE}$ to be *compressive* because it must balance the moment of the 300-N force about C.

EXAMPLE 6.6

Determine the force in member *CF* of the truss shown in Fig. 6–17*a*. Indicate whether the member is in tension or compression. Assume each member is pin connected.

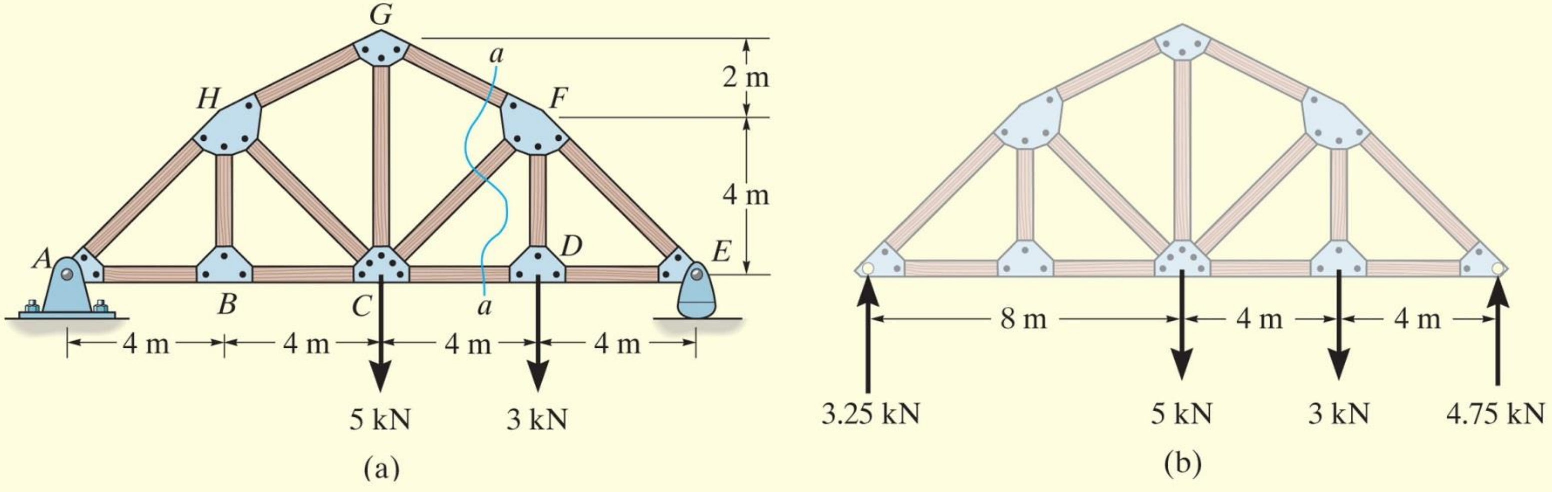


Fig. 6–17

EXAMPLE 6.6 CONTINUED

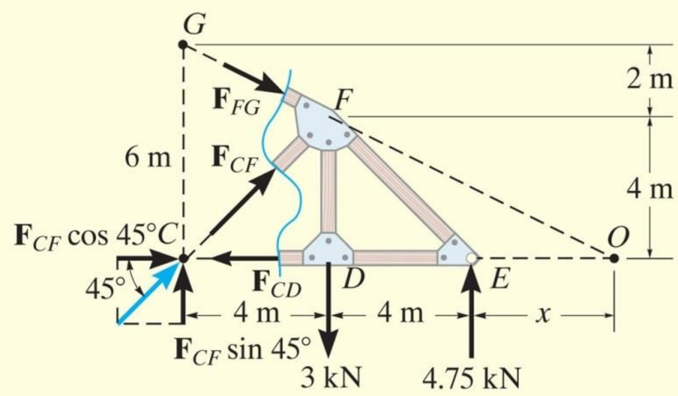
SOLUTION

Free-Body Diagram. Section *aa* in Fig. 6–17*a* will be used since this section will "expose" the internal force in member *CF* as "external" on the free-body diagram of either the right or left portion of the truss. It is first necessary, however, to determine the support reactions on either the left or right side. Verify the results shown on the free-body diagram in Fig. 6–17*b*.

The free-body diagram of the right portion of the truss, which is the easiest to analyze, is shown in Fig. 6–17*c*. There are three unknowns, F_{FG}, F_{CF}, and F_{CD}.

EXAMPLE 6.6 CONTINUED

Equations of Equilibrium. We will apply the moment equation about point O in order to eliminate the two unknowns F_{FG} and F_{CD}. The location of point O measured from E can be determined from proportional triangles, i.e., $4/(4+x) = 6/(8+x)$, $x = 4$ m. Or, stated in another manner, the slope of member GF has a drop of 2 m to a horizontal distance of 4 m. Since FD is 4 m, Fig. 6–17c, then from D to O the distance must be 8 m.



(c)

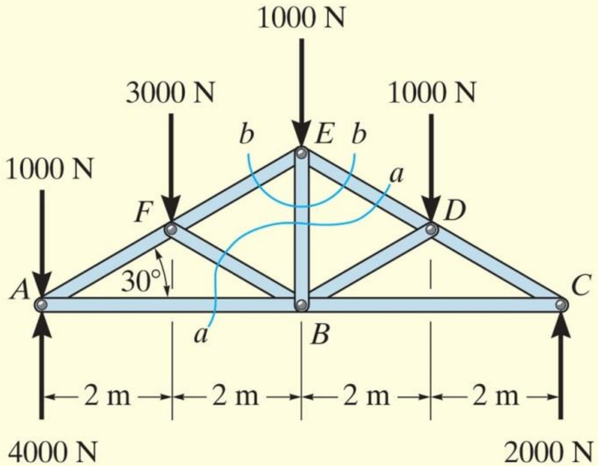
An easy way to determine the moment of $\mathbf{F}_{CF}$ about point O is to use the principle of transmissibility and slide $\mathbf{F}_{CF}$ to point C, and then resolve $\mathbf{F}_{CF}$ into its two rectangular components. We have

$$\circlearrowleft + \Sigma M_O = 0;$$

$$-F_{CF} \sin 45°(12 \text{ m}) + (3 \text{ kN})(8 \text{ m}) - (4.75 \text{ kN})(4 \text{ m}) = 0$$

$$F_{CF} = 0.589 \text{ kN} \quad \text{(C)} \qquad \textit{Ans.}$$

EXAMPLE 6.7



(a)

Determine the force in member *EB* of the roof truss shown in Fig. 6–18*a*. Indicate whether the member is in tension or compression.

SOLUTION

Free-Body Diagrams. By the method of sections, any imaginary section that cuts through *EB*, Fig. 6–18*a*, will also have to cut through three other members for which the forces are unknown. For example, section *aa* cuts through *ED*, *EB*, *FB*, and *AB*. If a free-body diagram of the left side of this section is considered, Fig. 6–18*b*, it is possible to obtain $\mathbf{F}_{ED}$ by summing moments about *B* to eliminate the other three unknowns; however, $\mathbf{F}_{EB}$ cannot be determined from the remaining two equilibrium equations. One possible way of obtaining $\mathbf{F}_{EB}$ is first to determine $\mathbf{F}_{ED}$ from section *aa*, then use this result on section *bb*, Fig. 6–18*a*, which is shown in Fig. 6–18*c*. Here the force system is concurrent and our sectioned free-body diagram is the same as the free-body diagram for the joint at *E*.

EXAMPLE 6.7 CONTINUED

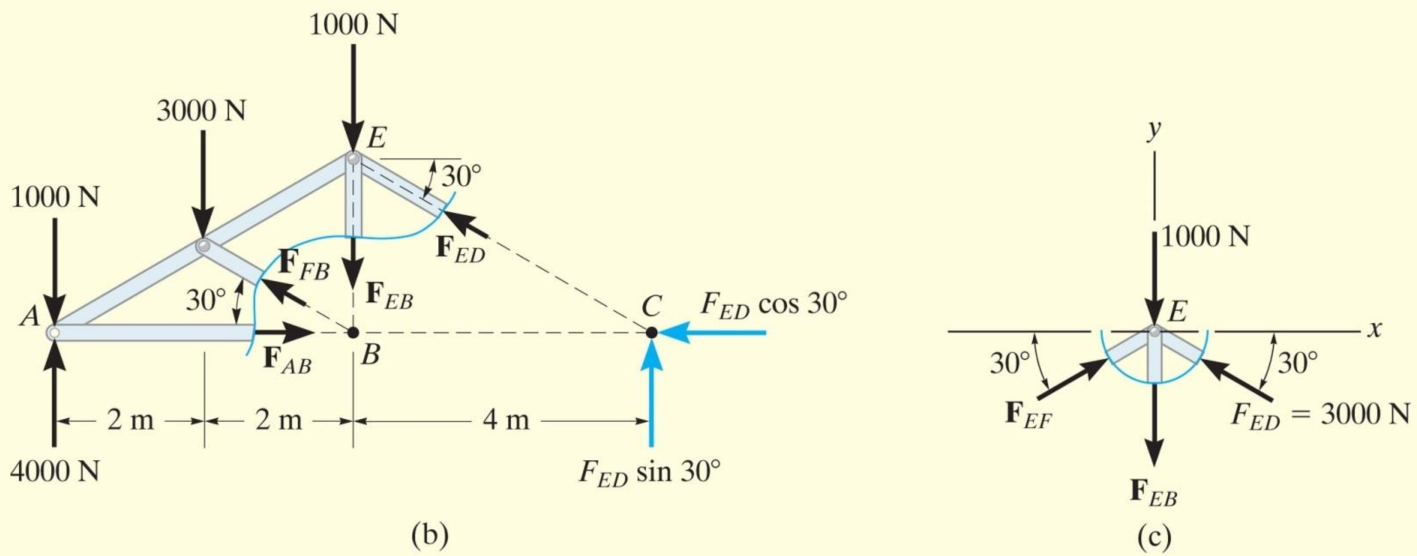


Fig. 6–18

EXAMPLE 6.7 CONTINUED

Equations of Equilibrium. In order to determine the moment of $\mathbf{F}_{ED}$ about point B, Fig. 6–18b, we will use the principle of transmissibility and slide the force to point C and then resolve it into its rectangular components as shown. Therefore,

$$\circlearrowleft + \Sigma M_B = 0; \qquad 1000 \text{ N}(4 \text{ m}) + 3000 \text{ N}(2 \text{ m}) - 4000 \text{ N}(4 \text{ m}) + F_{ED} \sin 30°(4 \text{ m}) = 0$$

$$F_{ED} = 3000 \text{ N} \quad \text{(C)}$$

Considering now the free-body diagram of section bb, Fig. 6–18c, we have

$$\xrightarrow{+} \Sigma F_x = 0; \qquad F_{EF} \cos 30° - 3000 \cos 30° \text{ N} = 0$$

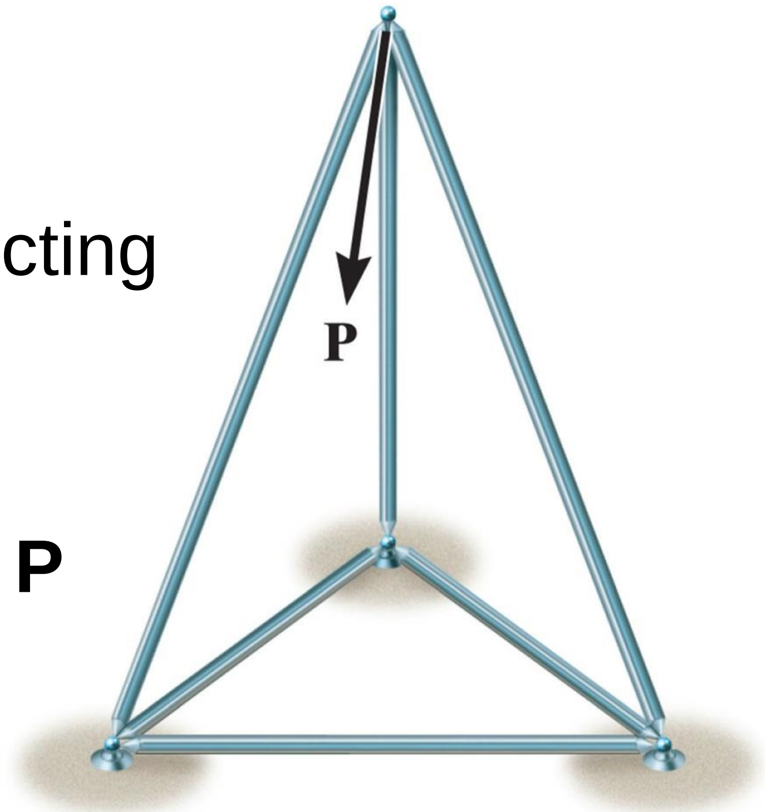
$$F_{EF} = 3000 \text{ N} \quad \text{(C)}$$

$$+\uparrow \Sigma F_y = 0; \qquad 2(3000 \sin 30° \text{ N}) - 1000 \text{ N} - F_{EB} = 0$$

$$F_{EB} = 2000 \text{ N} \quad \text{(T)} \qquad \textit{Ans.}$$

6.5 Space Trusses

- A space truss consists of members joined together at their ends to form 3D structure
- The simplest space truss is a tetrahedron, formed by connecting six members
- Additional members would be redundant in supporting force **P**



6.5 Space Trusses

Assumptions for Design

- ✓ Members of a space truss is treated as 2 force members provided the external loading is applied at the joints and the joints consist of ball-and-socket connection.
- ✓ When weight of the member is considered, apply it as a vertical force, half of its magnitude applied at each end of the member.

Figure: 06_PH005

Typical roof-supporting space truss. Notice the use of ball-and- socket joints for the connections.

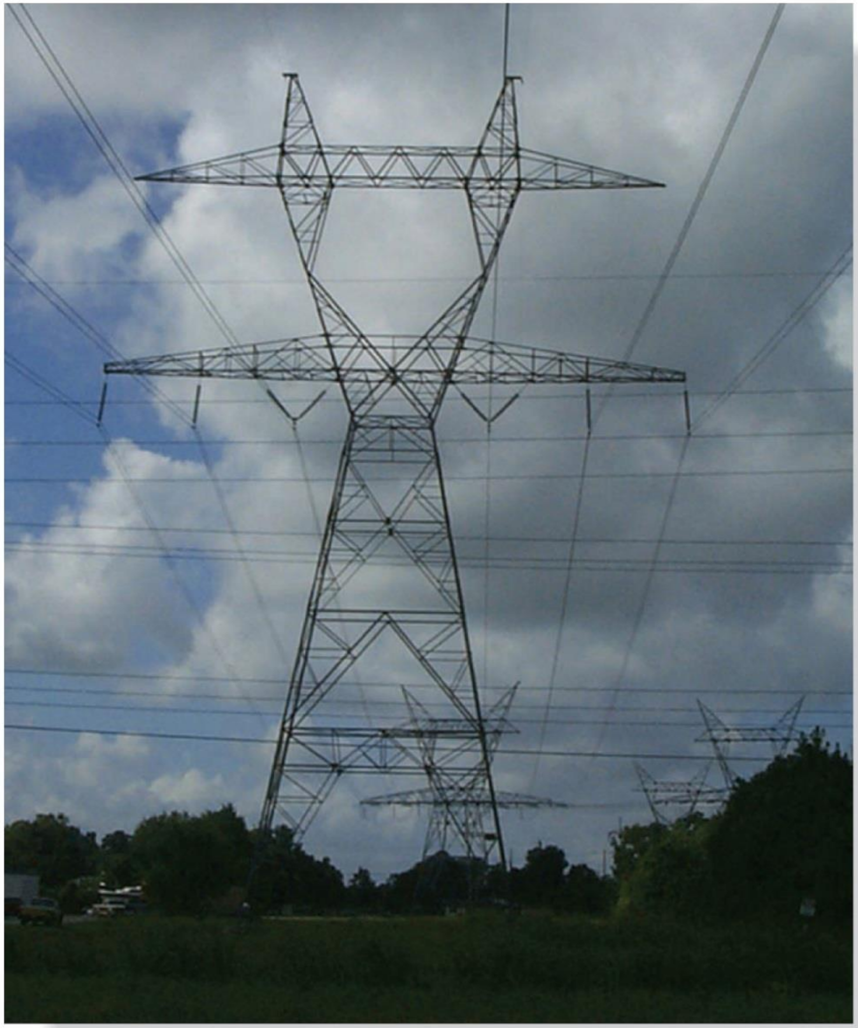


Figure: 06_PH006

For economic reasons, large electrical transmission towers are often constructed using space trusses.

Procedure for analysis

Either the method of joints or the method of sections can be used to determine the forces developed in the members of a simple space truss.

Method of Joints.

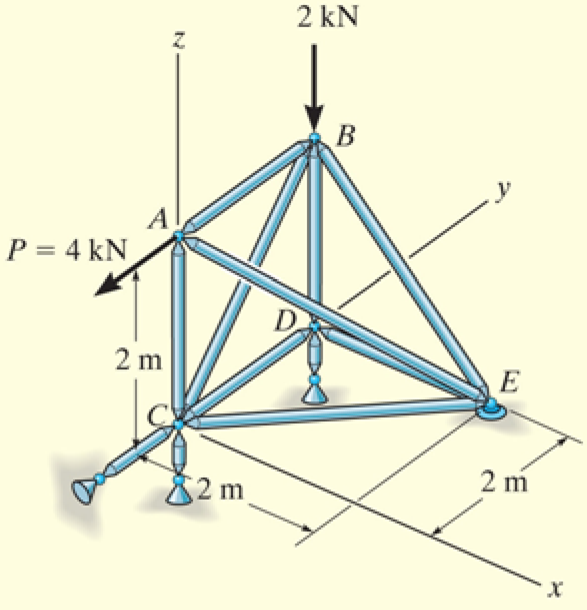
If the forces in *all* the members of the truss are to be determined, then the method of joints is most suitable for the analysis. Here it is necessary to apply the three equilibrium equations $\Sigma F_x = 0$, $\Sigma F_y = 0$, $\Sigma F_z = 0$ to the forces acting at each joint. Remember that the solution of many simultaneous equations can be avoided if the force analysis begins at a joint having at least one known force and at most three unknown forces. Also, if the three-dimensional geometry of the force system at the joint is hard to visualize, it is recommended that a Cartesian vector analysis be used for the solution.

Procedure for analysis

Method of Sections.

If only a *few* member forces are to be determined, the method of sections can be used. When an imaginary section is passed through a truss and the truss is separated into two parts, the force system acting on one of the segments must satisfy the *six* equilibrium equations: $\Sigma F_x = 0$, $\Sigma F_y = 0$, $\Sigma F_z = 0$, $\Sigma M_x = 0$, $\Sigma M_y = 0$, $\Sigma M_z = 0$ (Eqs. 5–6). By proper choice of the section and axes for summing forces and moments, many of the unknown member forces in a space truss can be computed *directly*, using a single equilibrium equation.

EXAMPLE 6.8



(a)

Determine the forces acting in the members of the space truss shown in Fig. 6–20*a*. Indicate whether the members are in tension or compression.

SOLUTION

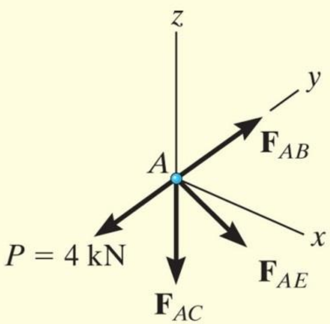
Since there are one known force and three unknown forces acting at joint *A*, the force analysis of the truss will begin at this joint.

Joint A. (Fig. 6–20*b*). Expressing each force acting on the free-body diagram of joint *A* as a Cartesian vector, we have

$$\mathbf{P} = \{-4\mathbf{j}\}\ \text{kN}, \qquad \mathbf{F}_{AB} = F_{AB}\,\mathbf{j}, \quad \mathbf{F}_{AC} = -F_{AC}\,\mathbf{k},$$

$$\mathbf{F}_{AE} = F_{AE}\left(\frac{\mathbf{r}_{AE}}{r_{AE}}\right) = F_{AE}(0.577\mathbf{i} + 0.577\mathbf{j} - 0.577\mathbf{k})$$

EXAMPLE 6.8 CONTINUED



(b)

For equilibrium,

$$\Sigma \mathbf{F} = \mathbf{0}; \qquad \mathbf{P} + \mathbf{F}_{AB} + \mathbf{F}_{AC} + \mathbf{F}_{AE} = \mathbf{0}$$

$$-4\mathbf{j} + F_{AB}\mathbf{j} - F_{AC}\mathbf{k} + 0.577F_{AE}\mathbf{i} + 0.577F_{AE}\mathbf{j} - 0.577F_{AE}\mathbf{k} = \mathbf{0}$$

$$\Sigma F_x = 0; \qquad 0.577F_{AE} = 0$$

$$\Sigma F_y = 0; \qquad -4 + F_{AB} + 0.577F_{AE} = 0$$

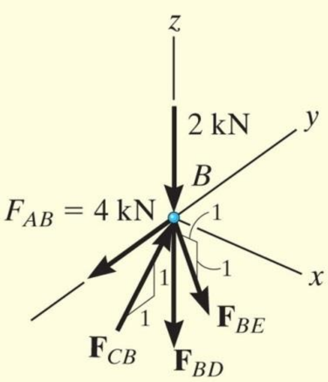
$$\Sigma F_z = 0; \qquad -F_{AC} - 0.577F_{AE} = 0$$

$$F_{AC} = F_{AE} = 0 \qquad \textit{Ans.}$$

$$F_{AB} = 4 \text{ kN} \quad \text{(T)} \qquad \textit{Ans.}$$

Since F_{AB} is known, joint B can be analyzed next.

EXAMPLE 6.8 CONTINUED



(c)

Fig. 6–20

Joint *B*. (Fig. 6–20*c*).

$$\Sigma F_x = 0; \qquad F_{BE}\frac{1}{\sqrt{2}} = 0$$

$$\Sigma F_y = 0; \qquad -4 + F_{CB}\frac{1}{\sqrt{2}} = 0$$

$$\Sigma F_z = 0; \qquad -2 + F_{BD} - F_{BE}\frac{1}{\sqrt{2}} + F_{CB}\frac{1}{\sqrt{2}} = 0$$

$$F_{BE} = 0, \qquad F_{CB} = 5.65 \text{ kN (C)} \qquad F_{BD} = 2 \text{ kN (T)} \qquad \textit{Ans.}$$

The *scalar* equations of equilibrium can now be applied to the forces acting on the free-body diagrams of joints *D* and *C*. Show that

$$F_{DE} = F_{DC} = F_{CE} = 0 \qquad \textit{Ans.}$$

6.6 Frames and Machines

Frames and Machines

- ✓ Composed of pin-connected multi-force members
- ✓ Frames → stationary, to support loads
- ✓ Machines→ moving parts
- ✓ Apply equations of equilibrium to each member to determine the unknown forces

6.6 Frames

This crane is a typical example of a framework.

Common tools such as these pliers act as simple machines. Here the applied force on the handles creates a much larger force at the jaws.

6.6 Frames and Machines

Free-Body Diagrams

- Isolate each part by drawing its outlined shape
  - Show all forces and/or couple moments act on the part
  - Identify each known and unknown force and couple moment
  - Indicate any dimension
  - Apply equations of equilibrium
  - Assumed sense of unknown force or couple moment

6.6 Frames and Machines

Free-Body Diagrams

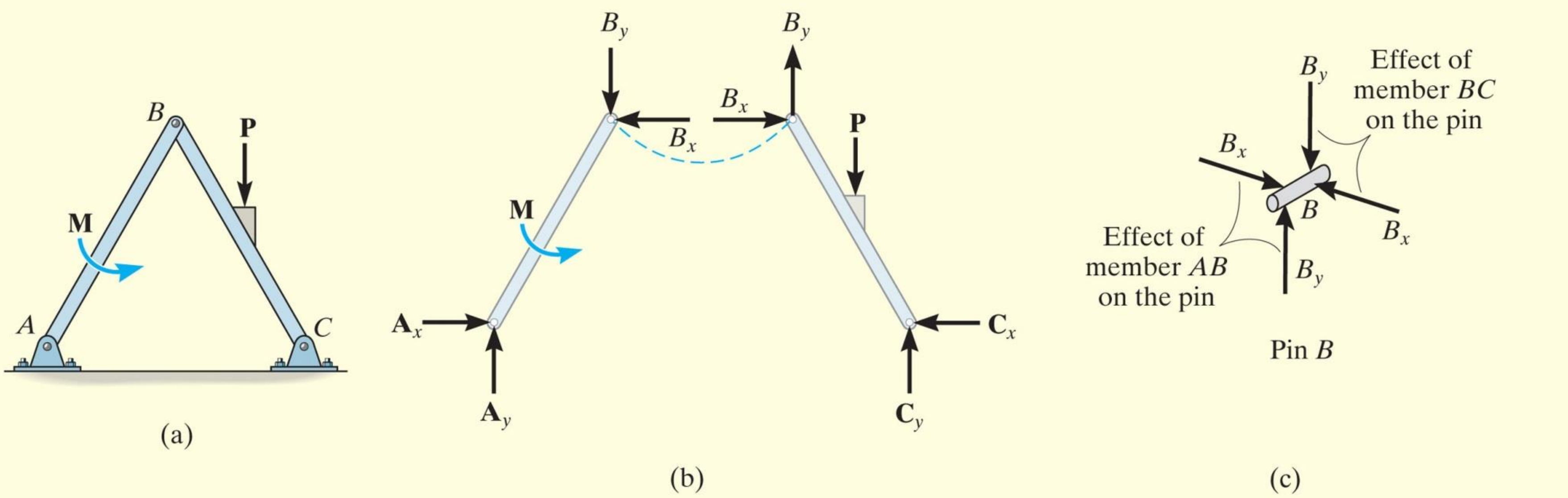
- Identify all the two force members in the structure and represent their FBD as having two equal but opposite collinear forces acting at their points of application

Free-Body Diagrams

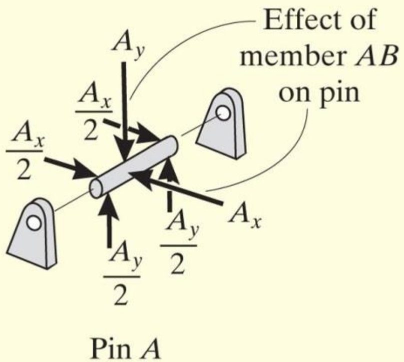
- Forces common to any two contracting member act with equal magnitudes but opposite sense on the respective members
  - - if the two members are treated as a “system” of connected members , these forces are “internal” and are not shown on the FBD
  - - if the FBD of each member is drawn, the forces are “external” and must be shown on the FBD

EXAMPLE 6.9

For the frame shown in Fig. 6–21*a*, draw the free-body diagram of (a) each member, (b) the pins at *B* and *A*, and (c) the two members connected together.



EXAMPLE 6.9 CONTINUED

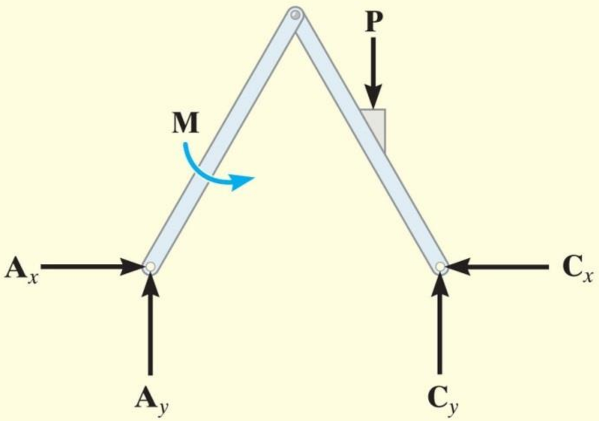


(d)

SOLUTION

Part (a). By inspection, members BA and BC are *not* two-force members. Instead, as shown on the free-body diagrams, Fig. 6–21b, BC is subjected to a force from each of the pins at B and C and the external force $\mathbf{P}$. Likewise, AB is subjected to a force from each of the pins at A and B and the external couple moment $\mathbf{M}$. The pin forces are represented by their x and y components.

EXAMPLE 6.9 CONTINUED



(e)

Fig. 6–21

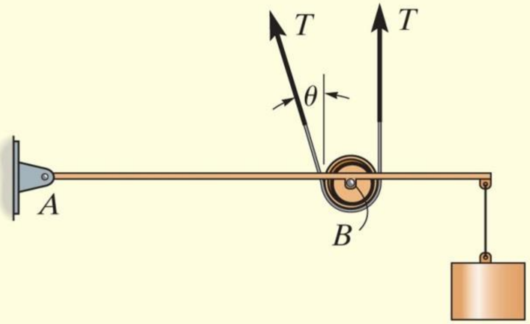
Part (b). The pin at B is subjected to only *two forces*, i.e., the force of member BC and the force of member AB. For *equilibrium* these forces (or their respective components) must be equal but opposite, Fig. 6–21*c*. Realize that Newton's third law is applied between the pin and its connected members, i.e., the effect of the pin on the two members, Fig. 6–21*b*, and the equal but opposite effect of the two members on the pin, Fig. 6–21*c*. In the same manner, there are three forces on pin A, Fig. 6–21*d*, caused by the force components of member AB and each of the two pin leafs.

Part (c). The free-body diagram of both members connected together, yet removed from the supporting pins at A and C, is shown in Fig. 6–21*e*. The force components $\mathbf{B}_x$ and $\mathbf{B}_y$ are *not shown* on this diagram since they are *internal* forces (Fig. 6–21*b*) and therefore cancel out. Also, to be consistent when later applying the equilibrium equations, the unknown force components at A and C must act in the *same sense* as those shown in Fig. 6–21*b*.

EXAMPLE 6.10

A constant tension in the conveyor belt is maintained by using the device shown in Fig. 6–22*a*. Draw the free-body diagrams of the frame and the cylinder (or pulley) that the belt surrounds. The suspended block has a weight of *W*.

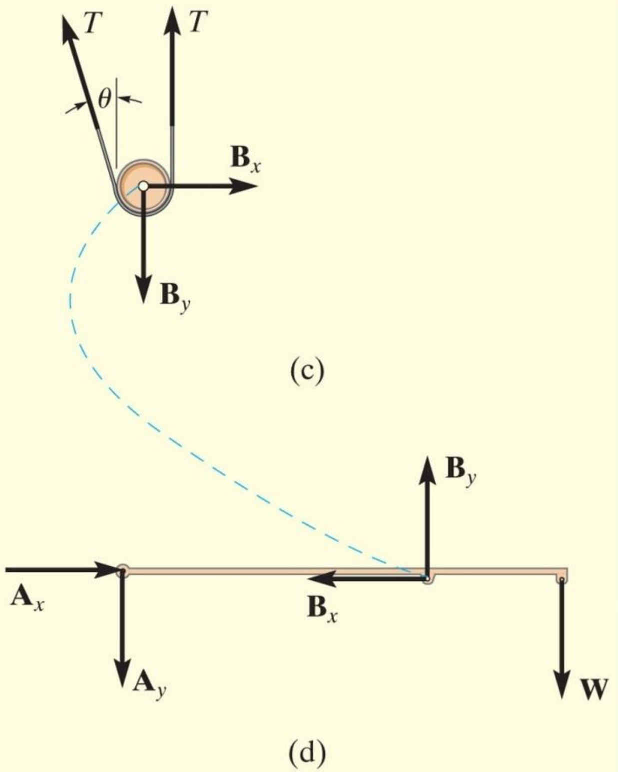
(a)



(b)

Fig. 6–22 (© Russell C. Hibbeler)

EXAMPLE 6.10 CONTINUED

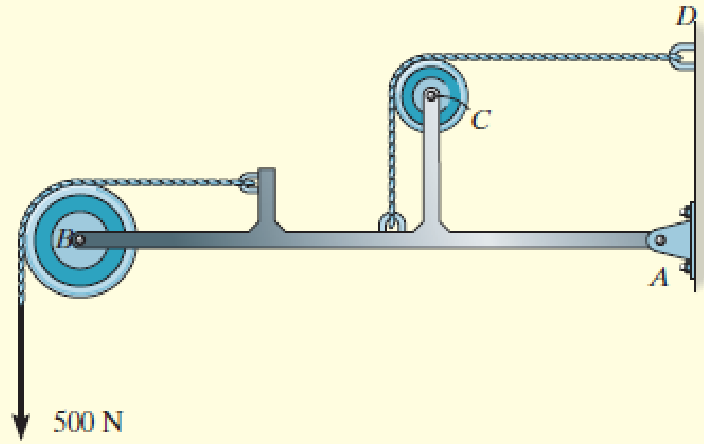


SOLUTION

The idealized model of the device is shown in Fig. 6–22*b*. Here the angle θ is assumed to be known. From this model, the free-body diagrams of the pulley and frame are shown in Figs. 6–22*c* and 6–22*d*, respectively. Note that the force components $\mathbf{B}_x$ and $\mathbf{B}_y$ that the pin at *B* exerts on the pulley must be equal but opposite to the ones acting on the frame. See Fig. 6–21*c* of Example 6.9.

EXAMPLE 6.11

For the frame shown in Fig. 6–23*a*, draw the free-body diagrams of (a) the entire frame including the pulleys and cords, (b) the frame without the pulleys and cords, and (c) each of the pulleys.



(a)

SOLUTION

Part (a). When the entire frame including the pulleys and cords is considered, the interactions at the points where the pulleys and cords are connected to the frame become pairs of *internal* forces which cancel each other and therefore are not shown on the free-body diagram, Fig. 6–23*b*.

EXAMPLE 6.11 CONTINUED

Part (b). When the cords and pulleys are removed, their effect *on the frame* must be shown, Fig. 6–23*c*.

Part (c). The force components B_x, B_y, C_x, C_y of the pins on the pulleys, Fig. 6–23*d*, are equal but opposite to the force components exerted by the pins on the frame, Fig. 6–23*c*. See Example 6.9.

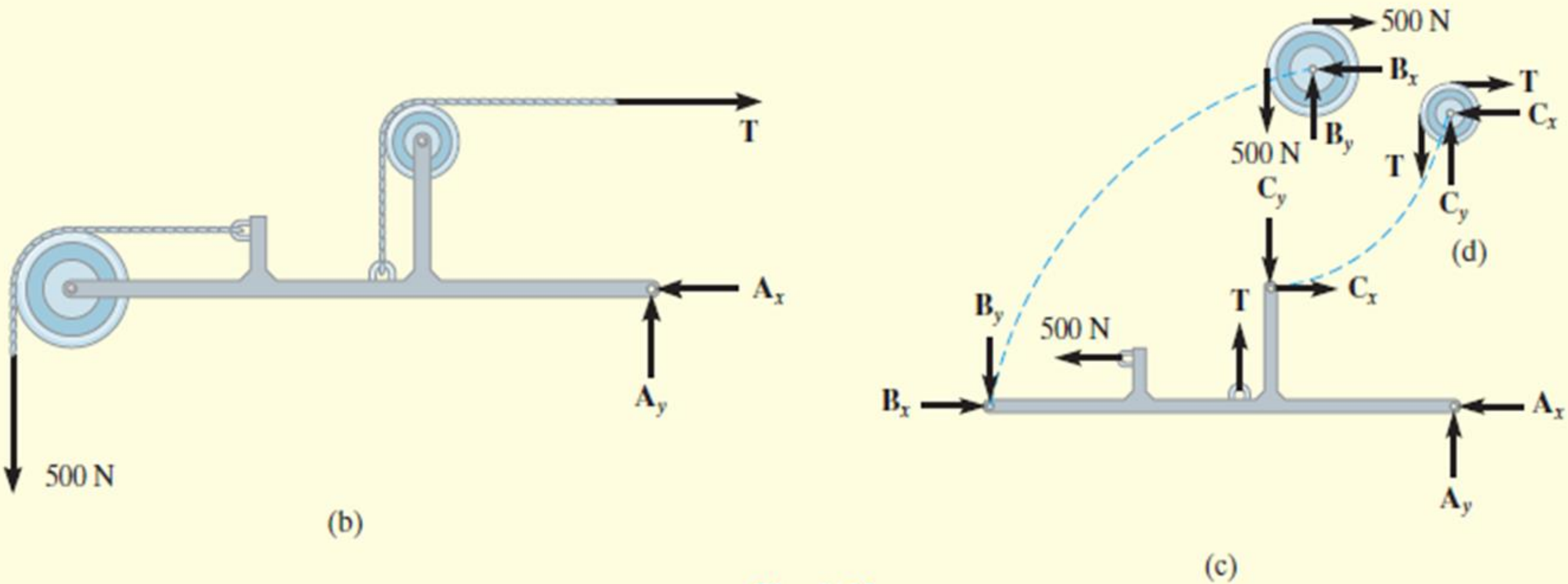


Fig. 6–23

EXAMPLE 6.12

Draw the free-body diagrams of the members of the backhoe, shown in the photo, Fig. 6–24*a*. The bucket and its contents have a weight W.

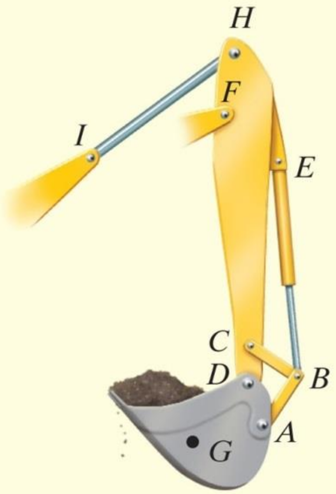
SOLUTION

The idealized model of the assembly is shown in Fig. 6–24*b*. By inspection, members AB, BC, BE, and HI are all two-force members since they are pin connected at their end points and no other forces act on them. The free-body diagrams of the bucket and the stick are shown in Fig. 6–24*c*. Note that pin C is subjected to only two forces, whereas the pin at B is subjected to three forces, Fig. 6–24*d*. The free-body diagram of the entire assembly is shown in Fig. 6–24*e*.

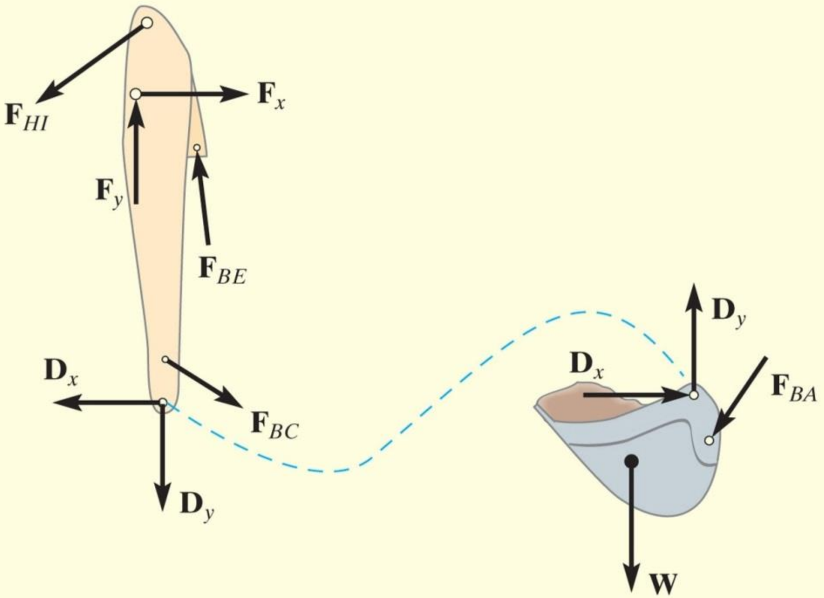
(a)

Fig. 6–24 (© Russell C. Hibbeler)

EXAMPLE 6.12 CONTINUED



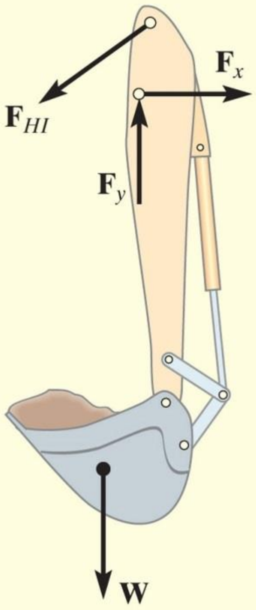
(b)



(c)

EXAMPLE 6.12 CONTINUED

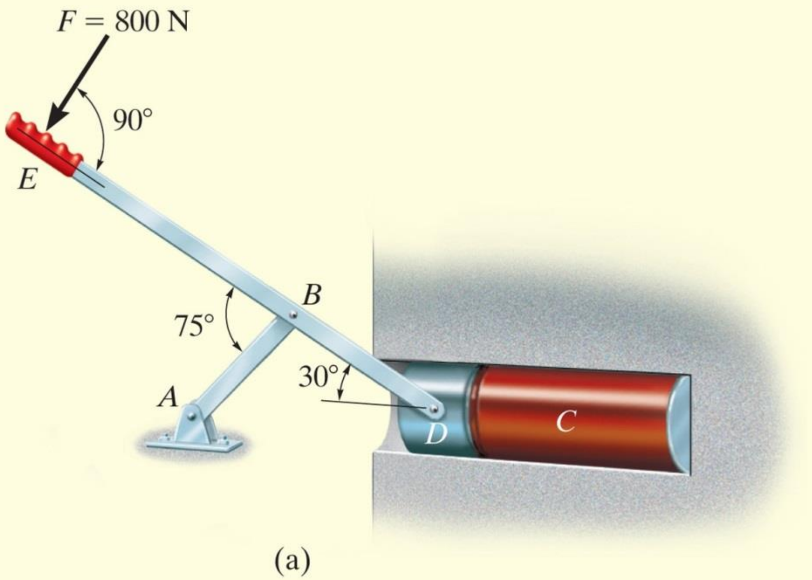
(d)



(e)

EXAMPLE 6.13

Draw the free-body diagram of each part of the smooth piston and link mechanism used to crush recycled cans, Fig. 6–25*a*.



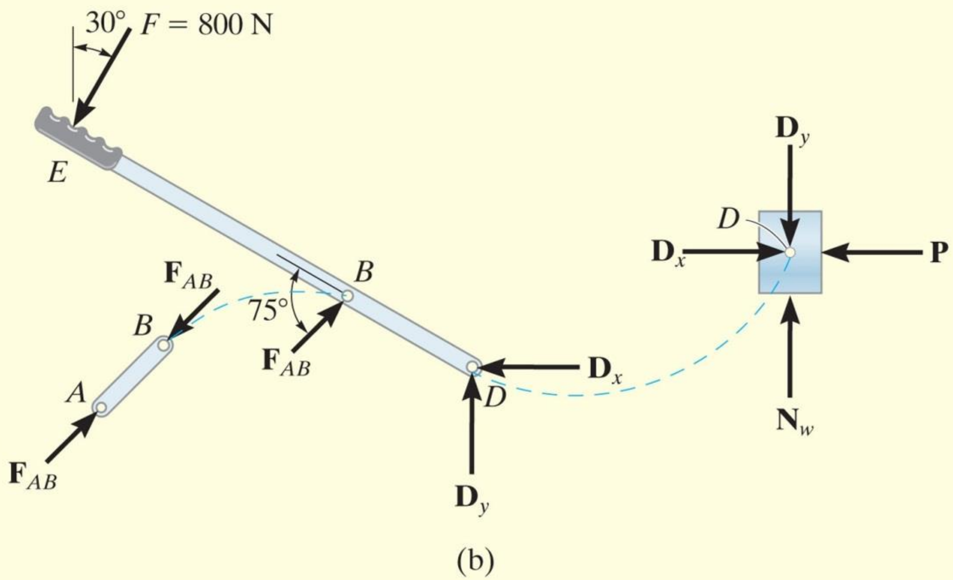
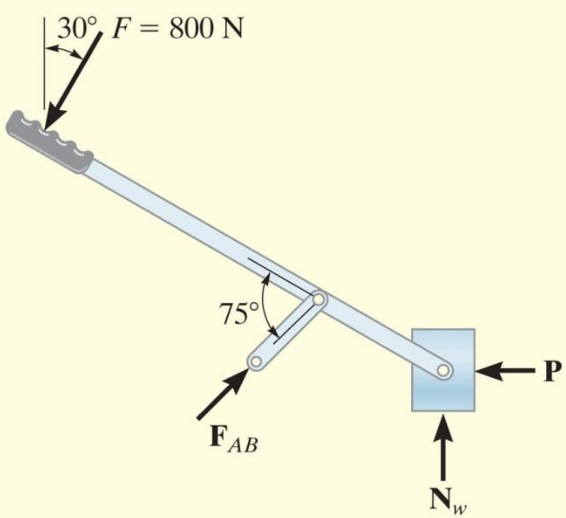


Fig. 6–25

EXAMPLE 6.13 CONTINUED



(c)

SOLUTION

By inspection, member *AB* is a two-force member. The free-body diagrams of the three parts are shown in Fig. 6–25*b*. Since the pins at *B* and *D* *connect only two parts together*, the forces there are shown as equal but opposite on the separate free-body diagrams of their connected members. In particular, four components of force act on the piston: $\mathbf{D}_x$ and $\mathbf{D}_y$ represent the effect of the pin (or lever *EBD*), $\mathbf{N}_w$ is the *resultant force* of the wall support, and **P** is the resultant compressive force caused by the can *C*. The directional sense of each of the unknown forces is assumed, and the correct sense will be established after the equations of equilibrium are applied.

NOTE: A free-body diagram of the entire assembly is shown in Fig. 6–25*c*. Here the forces between the components are internal and are not shown on the free-body diagram.

Procedure for analysis

The joint reactions on frames or machines (structures) composed of multiforce members can be determined using the following procedure.

Free-Body Diagram.

- Draw the free-body diagram of the entire frame or machine, a portion of it, or each of its members. The choice should be made so that it leads to the most direct solution of the problem.
- Identify the two-force members. Remember that regardless of their shape, they have equal but opposite collinear forces acting at their ends.
- When the free-body diagram of a group of members of a frame or machine is drawn, the forces between the connected parts of this group are internal forces and are not shown on the free-body diagram of the group.

Procedure for analysis

- Forces common to two members which are in contact act with equal magnitude but opposite sense on the respective free-body diagrams of the members.
- In many cases it is possible to tell by inspection the proper sense of the unknown forces acting on a member; however, if this seems difficult, the sense can be assumed.
- Remember that once the free-body diagram is drawn, a couple moment is a free vector and can act at any point on the diagram. Also, a force is a sliding vector and can act at any point along its line of action.

Procedure for analysis

Equations of Equilibrium.

- Count the number of unknowns and compare it to the total number of equilibrium equations that are available. In two dimensions, there are three equilibrium equations that can be written for each member.
- Sum moments about a point that lies at the intersection of the lines of action of as many of the unknown forces as possible.
- If the solution of a force or couple moment magnitude is found to be negative, it means the sense of the force is the reverse of that shown on the free-body diagram.

EXAMPLE 6.14

Determine the tension in the cables and also the force **P** required to support the 600-N force using the frictionless pulley system shown in Fig. 6–26*a*.

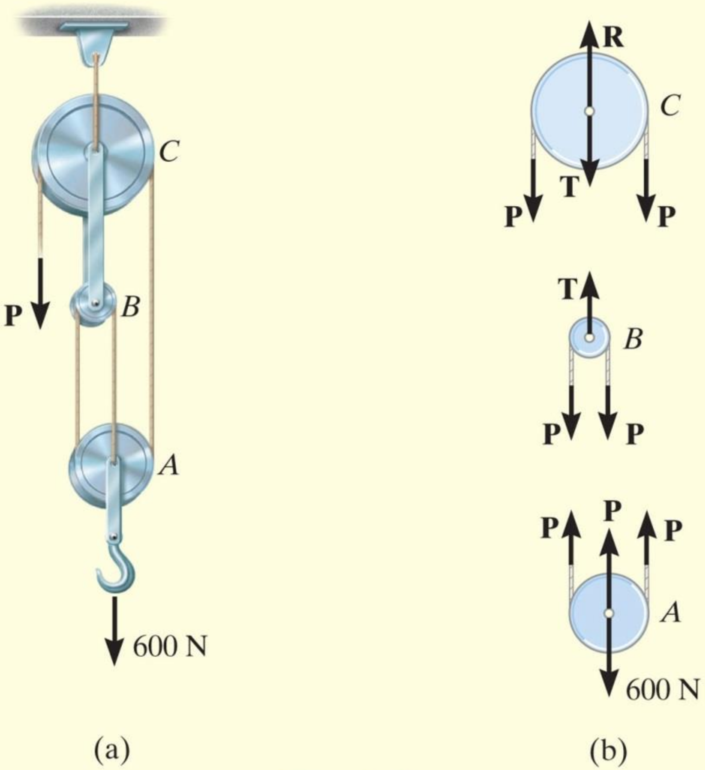


Fig. 6–26

EXAMPLE 6.14 CONTINUED

SOLUTION

Free-Body Diagram. A free-body diagram of each pulley *including* its pin and a portion of the contacting cable is shown in Fig. 6–26*b*. Since the cable is *continuous*, it has a *constant tension* P acting throughout its length. The link connection between pulleys B and C is a two-force member, and therefore it has an unknown tension T acting on it. Notice that the *principle of action, equal but opposite reaction* must be carefully observed for forces **P** and **T** when the *separate* free-body diagrams are drawn.

Equations of Equilibrium. The three unknowns are obtained as follows:

Pulley A

$$+\uparrow\Sigma F_y = 0; \qquad 3P - 600\text{ N} = 0 \qquad P = 200\text{ N} \qquad \textit{Ans.}$$

Pulley B

$$+\uparrow\Sigma F_y = 0; \qquad T - 2P = 0 \qquad T = 400\text{ N} \qquad \textit{Ans.}$$

Pulley C

$$+\uparrow\Sigma F_y = 0; \qquad R - 2P - T = 0 \qquad R = 800\text{ N} \qquad \textit{Ans.}$$

EXAMPLE 6.15

A 500-kg elevator car in Fig. 6–27*a* is being hoisted by motor *A* using the pulley system shown. If the car is traveling with a constant speed, determine the force developed in the two cables. Neglect the mass of the cable and pulleys.

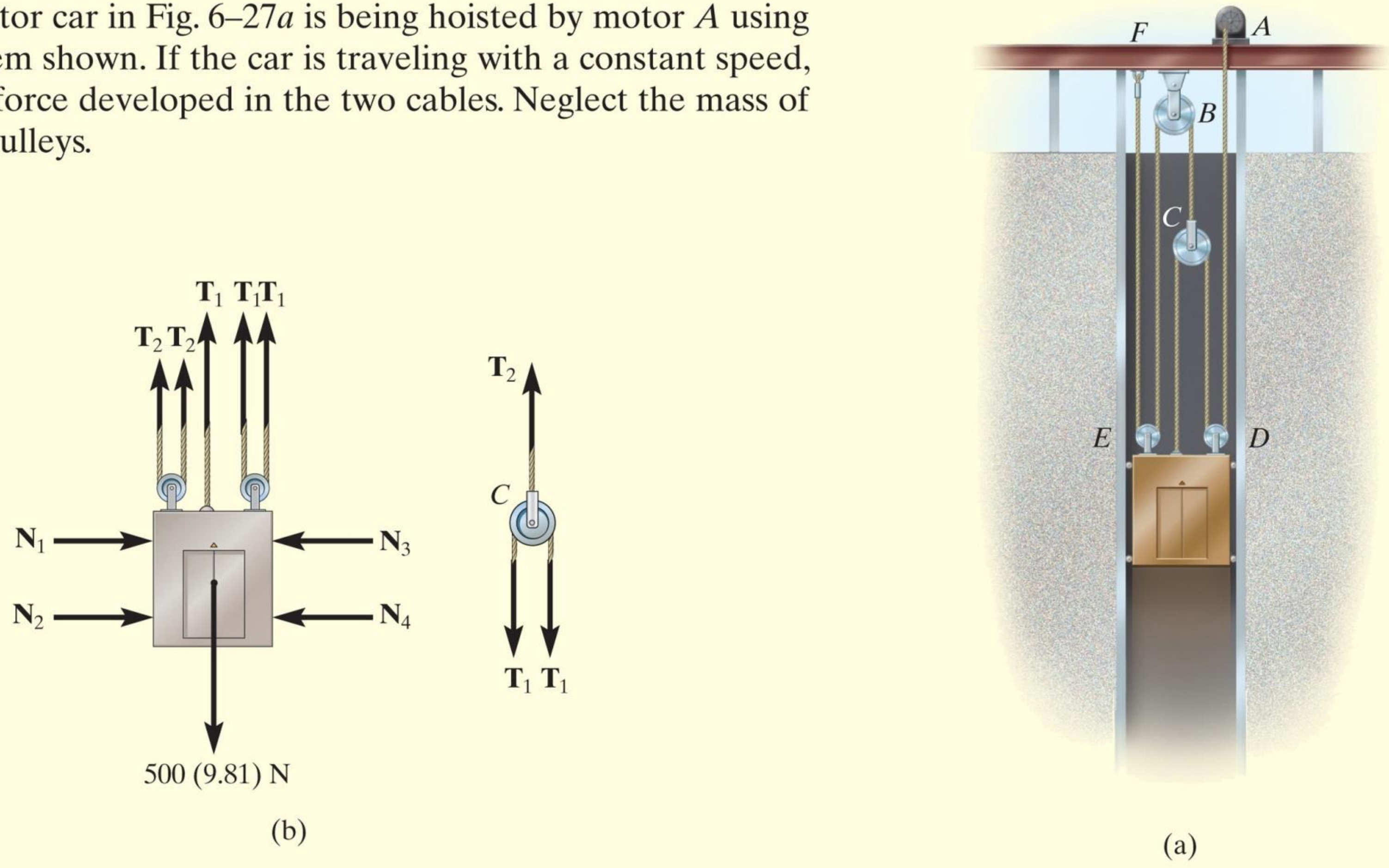


Fig. 6–27

EXAMPLE 6.15 CONTINUED

SOLUTION

Free-Body Diagram. We can solve this problem using the free-body diagrams of the elevator car and pulley C, Fig. 6–27b. The tensile forces developed in the cables are denoted as T_1 and T_2.

Equations of Equilibrium. For pulley C,

$$+\uparrow \Sigma F_y = 0; \qquad T_2 - 2T_1 = 0 \qquad \text{or} \qquad T_2 = 2T_1 \qquad (1)$$

For the elevator car,

$$+\uparrow \Sigma F_y = 0; \qquad 3T_1 + 2T_2 - 500(9.81)\text{ N} = 0 \qquad (2)$$

Substituting Eq. (1) into Eq. (2) yields

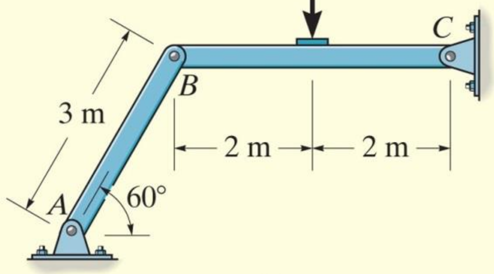
$$3T_1 + 2(2T_1) - 500(9.81)\text{ N} = 0$$

$$T_1 = 700.71\text{ N} = 701\text{ N} \qquad \textit{Ans.}$$

Substituting this result into Eq. (1),

$$T_2 = 2(700.71)\text{ N} = 1401\text{ N} = 1.40\text{ kN} \qquad \textit{Ans.}$$

EXAMPLE 6.16



(a)

Determine the horizontal and vertical components of force which the pin at C exerts on member BC of the frame in Fig. 6–28*a*.

SOLUTION I

Free-Body Diagrams. By inspection it can be seen that AB is a two-force member. The free-body diagrams are shown in Fig. 6–28*b*.

Equations of Equilibrium. The *three unknowns* can be determined by applying the three equations of equilibrium to member BC.

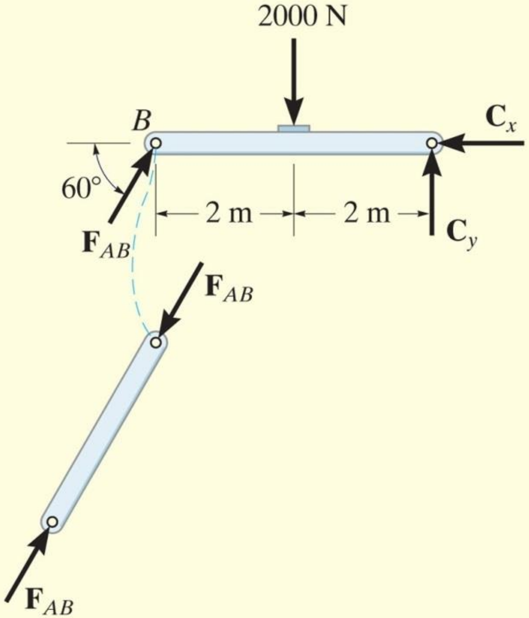
$$\circlearrowleft +\Sigma M_C = 0;\quad 2000\text{ N}(2\text{ m}) - (F_{AB}\sin 60^\circ)(4\text{ m}) = 0\quad F_{AB} = 1154.7\text{ N}$$

$$\xrightarrow{+} \Sigma F_x = 0;\quad 1154.7\cos 60^\circ\text{ N} - C_x = 0\quad C_x = 577\text{ N}\qquad \textit{Ans.}$$

$$+\uparrow \Sigma F_y = 0;\quad 1154.7\sin 60^\circ\text{ N} - 2000\text{ N} + C_y = 0$$

$$C_y = 1000\text{ N}\qquad \textit{Ans.}$$

EXAMPLE 6.16 CONTINUED



(b)

SOLUTION II

Free-Body Diagrams. If one does not recognize that AB is a two-force member, then more work is involved in solving this problem. The free-body diagrams are shown in Fig. 6–28*c*.

Equations of Equilibrium. The *six unknowns* are determined by applying the three equations of equilibrium to each member.

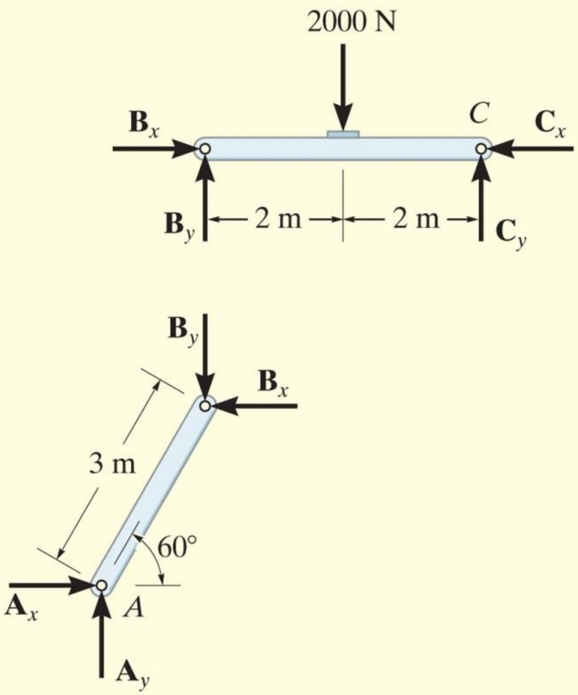
Member AB

$$\circlearrowleft + \Sigma M_A = 0; \quad B_x(3 \sin 60^\circ \text{ m}) - B_y(3 \cos 60^\circ \text{ m}) = 0 \qquad (1)$$

$$\xrightarrow{+} \Sigma F_x = 0; \quad A_x - B_x = 0 \qquad (2)$$

$$+\uparrow \Sigma F_y = 0; \quad A_y - B_y = 0 \qquad (3)$$

EXAMPLE 6.16 CONTINUED



(c)

Fig. 6–28

Member BC

$$\zeta+\Sigma M_C = 0; \quad 2000\text{ N}(2\text{ m}) - B_y(4\text{ m}) = 0 \tag{4}$$

$$\overset{+}{\rightarrow}\Sigma F_x = 0; \quad B_x - C_x = 0 \tag{5}$$

$$+\uparrow\Sigma F_y = 0; \quad B_y - 2000\text{ N} + C_y = 0 \tag{6}$$

The results for C_x and C_y can be determined by solving these equations in the following sequence: 4, 1, 5, then 6. The results are

$$B_y = 1000\text{ N}$$

$$B_x = 577\text{ N}$$

$$C_x = 577\text{ N} \qquad \textit{Ans.}$$

$$C_y = 1000\text{ N} \qquad \textit{Ans.}$$

By comparison, Solution I is simpler since the requirement that F_{AB} in Fig. 6–28*b* be equal, opposite, and collinear at the ends of member AB automatically satisfies Eqs. 1, 2, and 3 above and therefore eliminates the need to write these equations. ***As a result, save yourself some time and effort by always identifying the two-force members before starting the analysis!***

EXAMPLE 6.17

The compound beam shown in Fig. 6–29*a* is pin connected at *B*. Determine the components of reaction at its supports. Neglect its weight and thickness.

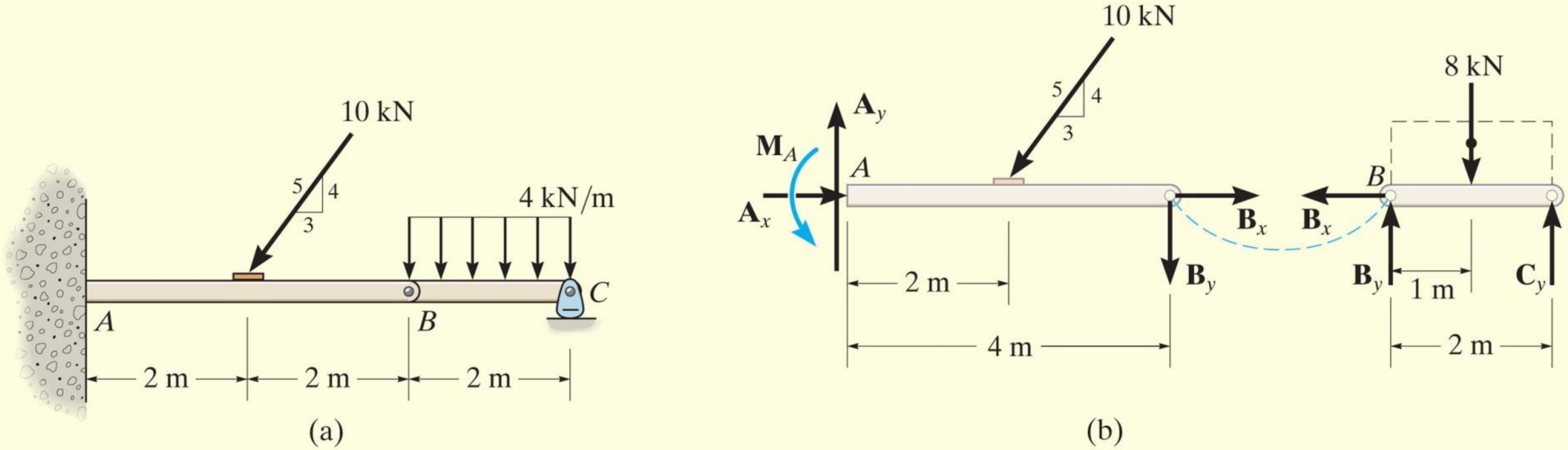


Fig. 6–29

SOLUTION

Free-Body Diagrams. By inspection, if we consider a free-body diagram of the *entire beam ABC*, there will be three unknown reactions at *A* and one at *C*. These four unknowns cannot all be obtained from the three available equations of equilibrium, and so for the solution it will become necessary to dismember the beam into its two segments, as shown in Fig. 6–29*b*.

EXAMPLE 6.17 CONTINUED

Equations of Equilibrium. The six unknowns are determined as follows:

Segment BC

$$\overset{+}{\leftarrow} \Sigma F_x = 0; \qquad B_x = 0$$

$$\circlearrowleft + \Sigma M_B = 0; \qquad -8\text{ kN}(1\text{ m}) + C_y(2\text{ m}) = 0$$

$$+\uparrow \Sigma F_y = 0; \qquad B_y - 8\text{ kN} + C_y = 0$$

Segment AB

$$\overset{+}{\rightarrow} \Sigma F_x = 0; \qquad A_x - (10\text{ kN})\left(\tfrac{3}{5}\right) + B_x = 0$$

$$\circlearrowleft + \Sigma M_A = 0; \qquad M_A - (10\text{ kN})\left(\tfrac{4}{5}\right)(2\text{ m}) - B_y(4\text{ m}) = 0$$

$$+\uparrow \Sigma F_y = 0; \qquad A_y - (10\text{ kN})\left(\tfrac{4}{5}\right) - B_y = 0$$

Solving each of these equations successively, using previously calculated results, we obtain

$$A_x = 6\text{ kN} \qquad A_y = 12\text{ kN} \qquad M_A = 32\text{ kN}\cdot\text{m} \qquad \textit{Ans.}$$

$$B_x = 0 \qquad B_y = 4\text{ kN}$$

$$C_y = 4\text{ kN} \qquad \textit{Ans.}$$

EXAMPLE 6.18

The two planks in Fig. 6–30*a* are connected together by cable *BC* and a smooth spacer *DE*. Determine the reactions at the smooth supports *A* and *F*, and also find the force developed in the cable and spacer.

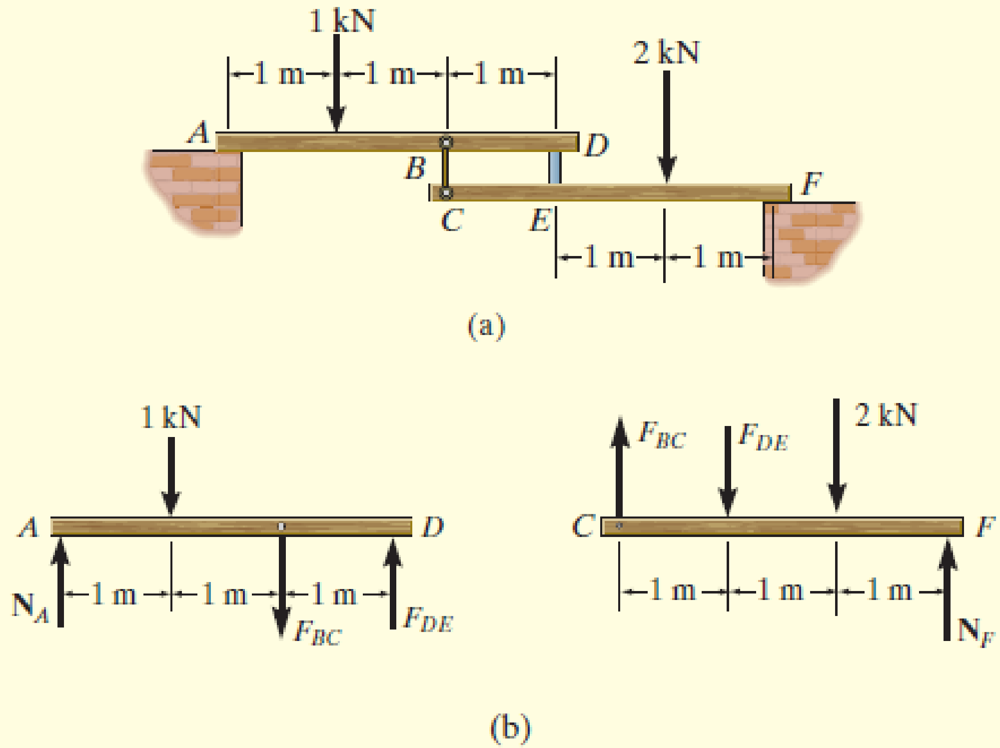


Fig. 6–30

EXAMPLE 6.18 CONTINUED

SOLUTION

Free-Body Diagrams. The free-body diagram of each plank is shown in Fig. 6–30*b*. It is important to apply Newton's third law to the interaction forces F_{BC} and F_{DE} as shown.

Equations of Equilibrium. For plank AD,

$$\circlearrowleft +\Sigma M_A = 0; \qquad F_{DE}(3\text{ m}) - F_{BC}(2\text{ m}) - 1\text{ kN}\,(1\text{ m}) = 0$$

For plank CF,

$$\circlearrowleft +\Sigma M_F = 0; \qquad F_{DE}(2\text{ m}) - F_{BC}(3\text{ m}) + 2\text{ kN}\,(1\text{ m}) = 0$$

Solving simultaneously,

$$F_{DE} = 1.40\text{ kN} \quad F_{BC} = 1.60\text{ kN} \qquad \textit{Ans.}$$

Using these results, for plank AD,

$$+\uparrow \Sigma F_y = 0; \qquad N_A + 1.40\text{ kN} - 1.60\text{ kN} - 1\text{ kN} = 0$$

$$N_A = 1.20\text{ kN} \qquad \textit{Ans.}$$

And for plank CF,

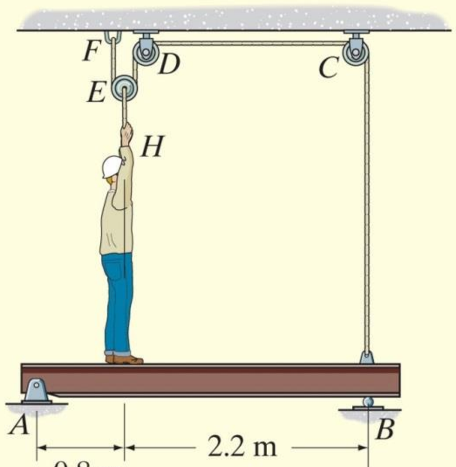
$$+\uparrow \Sigma F_y = 0; \qquad N_F + 1.60\text{ kN} - 1.40\text{ kN} - 2\text{ kN} = 0$$

$$N_F = 1.80\text{ kN} \qquad \textit{Ans.}$$

NOTE: Draw the free-body diagram of the system of *both* planks and apply $\Sigma M_A = 0$ to determine N_F. Then use the free-body diagram of CEF to determine F_{DE} and F_{BC}.

EXAMPLE 6.19

The 75-kg man in Fig. 6–31*a* attempts to lift the 40-kg uniform beam off the roller support at B. Determine the tension developed in the cable attached to B and the normal reaction of the man on the beam when this is about to occur.



(a)

SOLUTION

Free-Body Diagrams. The tensile force in the cable will be denoted as T_1. The free-body diagrams of the pulley E, the man, and the beam are shown in Fig. 6–31*b*. Since the man must lift the beam off the roller B then $N_B = 0$. When drawing each of these diagrams, it is very important to apply Newton's third law.

EXAMPLE 6.19 CONTINUED

Equations of Equilibrium. Using the free-body diagram of pulley E,

$$+\uparrow\Sigma F_y = 0; \quad 2T_1 - T_2 = 0 \quad \text{or} \quad T_2 = 2T_1 \qquad (1)$$

Referring to the free-body diagram of the man using this result,

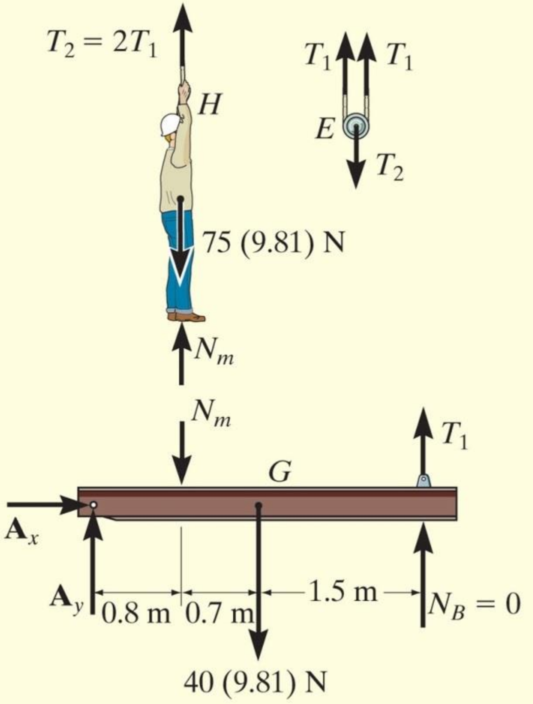
$$+\uparrow\Sigma F_y = 0 \quad N_m + 2T_1 - 75(9.81)\text{ N} = 0 \qquad (2)$$

Summing moments about point A on the beam,

$$\circlearrowleft+\Sigma M_A = 0; \; T_1(3\text{ m}) - N_m(0.8\text{ m}) - [40(9.81)\text{ N}]\,(1.5\text{ m}) = 0 \qquad (3)$$

Solving Eqs. 2 and 3 simultaneously for T_1 and N_m, then using Eq. (1) for T_2, we obtain

$$T_1 = 256\text{ N} \qquad N_m = 224\text{ N} \qquad T_2 = 512\text{ N} \qquad \textit{Ans.}$$



(b)

EXAMPLE 6.19 CONTINUED

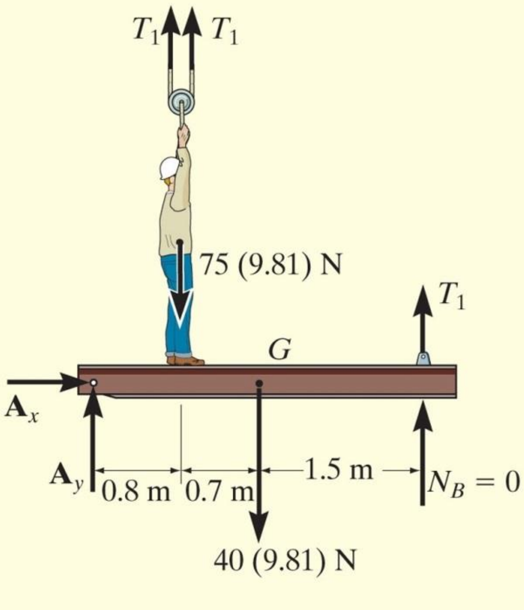
SOLUTION II

A direct solution for T_1 can be obtained by considering the beam, the man, and pulley E as a *single system*. The free-body diagram is shown in Fig. 6–31c. Thus,

$$\circlearrowleft +\Sigma M_A = 0; \qquad 2T_1(0.8\text{ m}) - [75(9.81)\text{ N}](0.8\text{ m}) - [40(9.81)\text{ N}](1.5\text{ m}) + T_1(3\text{ m}) = 0$$

$$T_1 = 256\text{ N} \qquad \textit{Ans.}$$

With this result Eqs. 1 and 2 can then be used to find N_m and T_2.

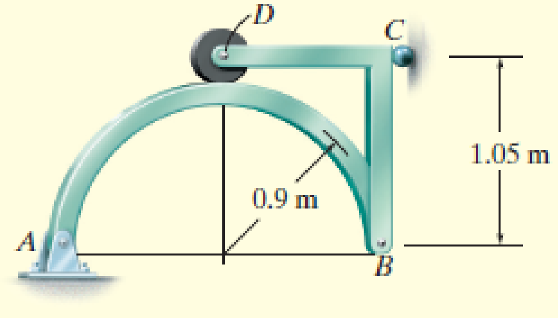


(c)

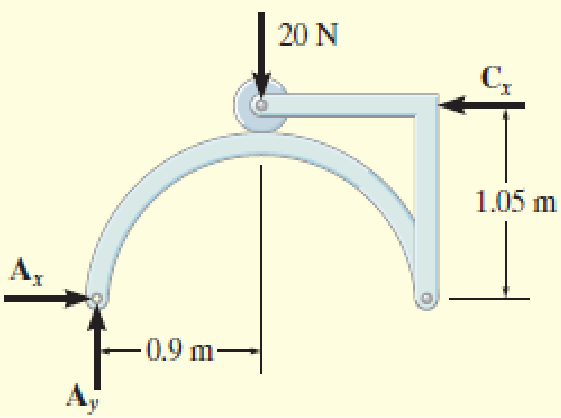
Fig. 6–31

EXAMPLE 6.20

The smooth disk shown in Fig. 6–32*a* is pinned at *D* and has a weight of 20 N. Neglecting the weights of the other members, determine the horizontal and vertical components of reaction at pins *B* and *D*.



(a)



SOLUTION

Free-Body Diagrams. The free-body diagrams of the entire frame and each of its members are shown in Fig. 6–32*b*.

EXAMPLE 6.20 CONTINUED

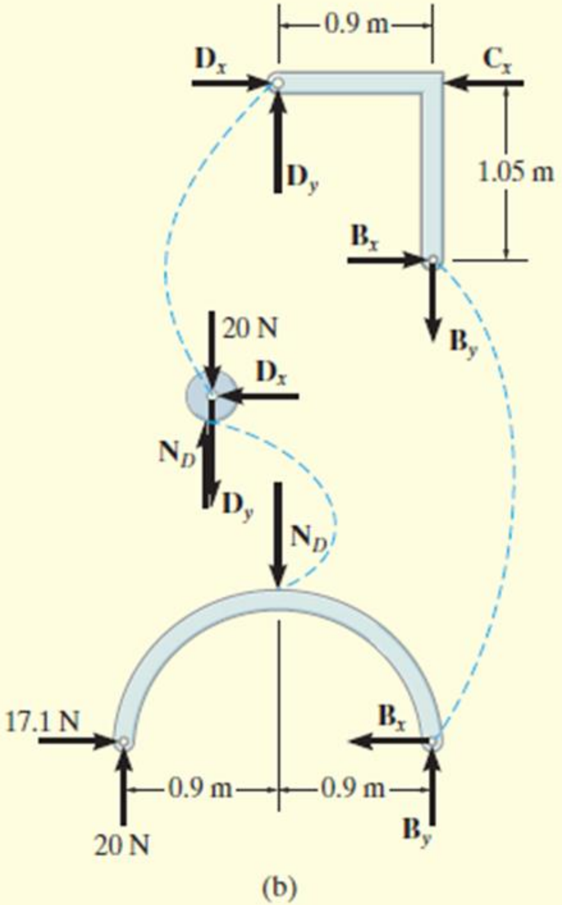


Fig. 6–32

Equations of Equilibrium. The eight unknowns can of course be obtained by applying the eight equilibrium equations to each member—three to member *AB*, three to member *BCD*, and two to the disk. (Moment equilibrium is automatically satisfied for the disk.) If this is done, however, all the results can be obtained only from a simultaneous solution of some of the equations. (Try it and find out.) To avoid this situation, it is best first to determine the three support reactions on the *entire* frame; then, using these results, the remaining five equilibrium equations can be applied to two other parts in order to solve successively for the other unknowns.

Entire Frame

$\circlearrowleft + \Sigma M_A = 0; \quad -(20\text{ N})(0.9\text{ m}) + C_x(1.05\text{ m}) = 0 \quad C_x = 17.1\text{ N}$

$\overset{+}{\rightarrow} \Sigma F_x = 0; \quad A_x - 17.1\text{ N} = 0 \quad A_x = 17.1\text{ N}$

$+\uparrow \Sigma F_y = 0; \quad A_y - 20\text{ N} = 0 \quad A_y = 20\text{ N}$

Member AB

$\overset{+}{\rightarrow} \Sigma F_x = 0; \quad 17.1\text{ N} - B_x = 0 \quad B_x = 17.1\text{ N}$ *Ans.*

$\circlearrowleft + \Sigma M_B = 0; \quad -(20\text{ N})(1.8\text{ m}) + N_D(0.9\text{ m}) = 0 \quad N_D = 40\text{ N}$

$+\uparrow \Sigma F_y = 0; \quad 20\text{ N} - 40\text{ N} + B_y = 0 \quad B_y = 20\text{ N}$ *Ans.*

Disk

$\overset{+}{\rightarrow} \Sigma F_x = 0; \quad D_x = 0$ *Ans.*

$+\uparrow \Sigma F_y = 0; \quad 40\text{ N} - 20\text{ N} - D_y = 0 \quad D_y = 20\text{ N}$ *Ans.*

EXAMPLE 6.21

The frame in Fig. 6–33*a* supports the 50-kg cylinder. Determine the horizontal and vertical components of reaction at *A* and the force at *C*.

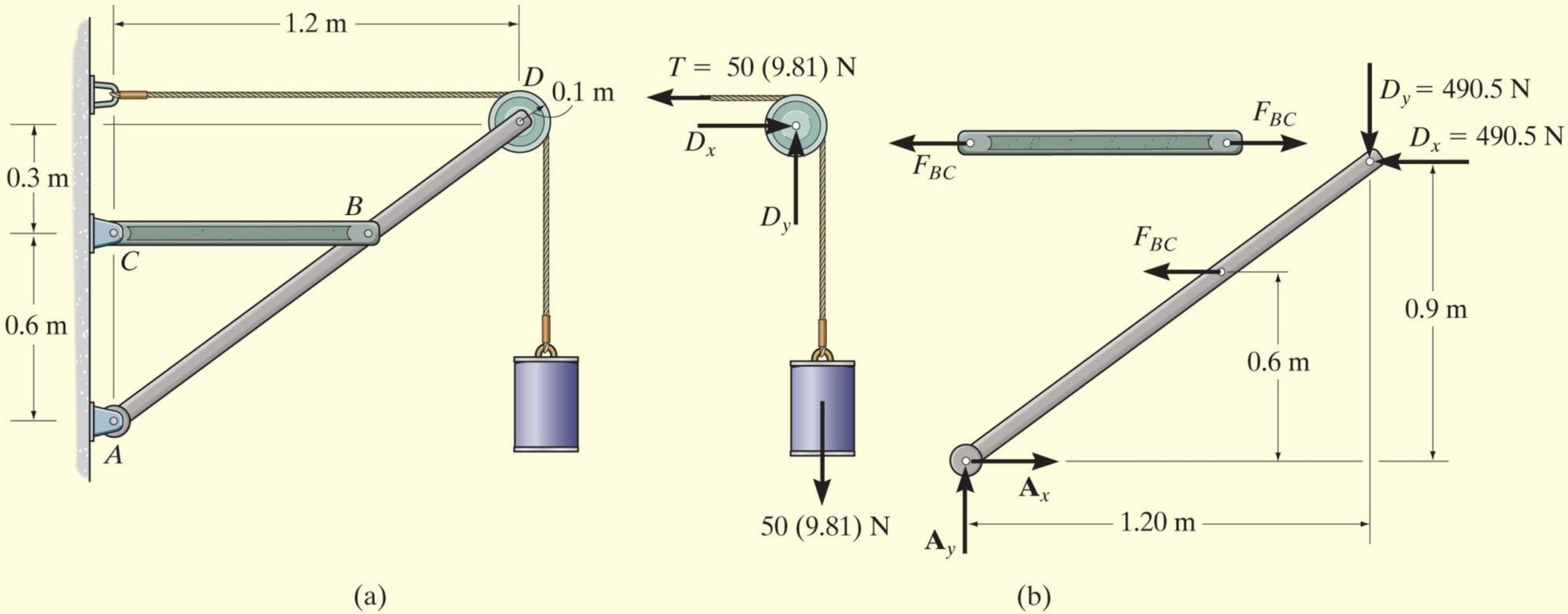


Fig. 6–33

EXAMPLE 6.21 CONTINUED

SOLUTION

Free-Body Diagrams. The free-body diagram of pulley D, along with the cylinder and a portion of the cord (a system), is shown in Fig. 6–33b. Member BC is a two-force member as indicated by its free-body diagram. The free-body diagram of member ABD is also shown.

Equations of Equilibrium. We will begin by analyzing the equilibrium of the pulley. The moment equation of equilibrium is automatically satisfied with $T = 50(9.81)$ N, and so

$$\overset{+}{\rightarrow}\Sigma F_x = 0; \qquad D_x - 50(9.81)\text{ N} = 0 \quad D_x = 490.5\text{ N}$$

$$+\uparrow\Sigma F_y = 0; \qquad D_y - 50(9.81)\text{ N} = 0 \quad D_y = 490.5\text{ N} \qquad \textit{Ans.}$$

Using these results, F_{BC} can be determined by summing moments about point A on member ABD.

$$\circlearrowleft +\Sigma M_A = 0; \; F_{BC}\,(0.6\text{ m}) + 490.5\text{ N}(0.9\text{ m}) - 490.5\text{ N}(1.20\text{ m}) = 0$$

$$F_{BC} = 245.25\text{ N} \qquad \textit{Ans.}$$

Now A_x and A_y can be determined by summing forces.

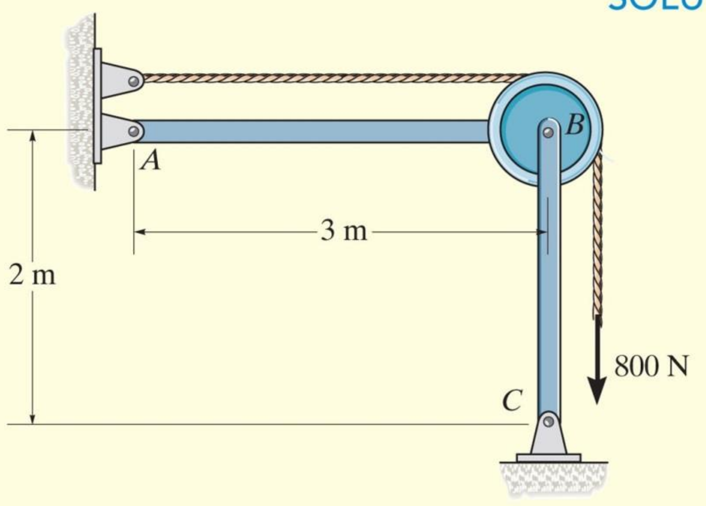
$$\overset{+}{\rightarrow}\Sigma F_x = 0; \quad A_x - 245.25\text{ N} - 490.5\text{ N} = 0 \quad A_x = 736\text{ N} \qquad \textit{Ans.}$$

$$+\uparrow\Sigma F_y = 0; \qquad A_y - 490.5\text{ N} = 0 \quad A_y = 490.5\text{ N} \qquad \textit{Ans.}$$

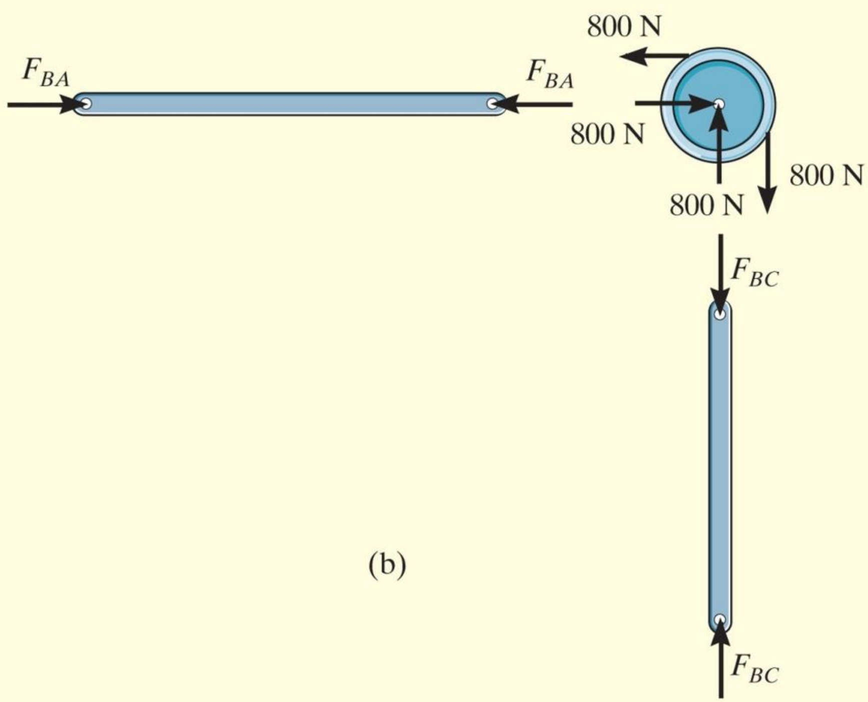
EXAMPLE 6.22

Determine the force the pins at A and B exert on the two-member frame shown in Fig. 6–34*a*.

SOLUTION I

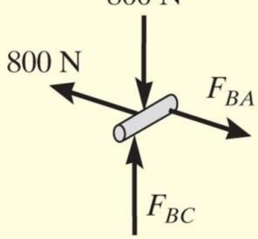


(a)



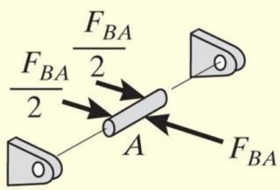
(b)

EXAMPLE 6.22 CONTINUED



Pin B

(c)



Pin A

(d)

Free-Body Diagrams. By inspection AB and BC are two-force members. Their free-body diagrams, along with that of the pulley, are shown in Fig. 6–34*b*. In order to solve this problem we must also include the free-body diagram of the pin at B because this pin connects all *three members* together, Fig. 6–34*c*.

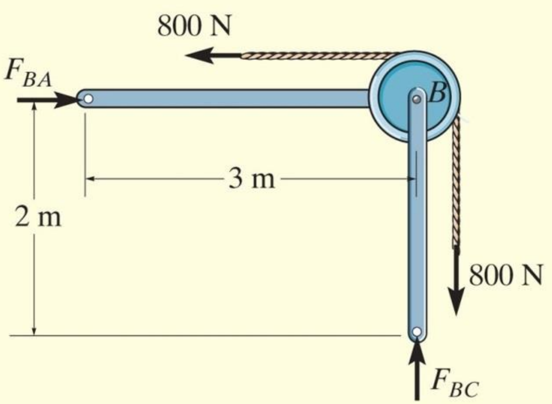
Equations of Equilibrium: Apply the equations of force equilibrium to pin B.

$$\xrightarrow{+}\Sigma F_x = 0; \qquad F_{BA} - 800\text{ N} = 0; \qquad F_{BA} = 800\text{ N} \qquad \textit{Ans.}$$

$$+\uparrow\Sigma F_y = 0; \qquad F_{BC} - 800\text{ N} = 0; \qquad F_{BC} = 800\text{ N} \qquad \textit{Ans.}$$

NOTE: The free-body diagram of the pin at A, Fig. 6–34*d*, indicates how the force F_{AB} is balanced by the force $(F_{AB}/2)$ exerted on the pin by each of the two pin leaves.

EXAMPLE 6.22 CONTINUED



(e)

Fig. 6–34

SOLUTION II

Free-Body Diagram. If we realize that AB and BC are two-force members, then the free-body diagram of the *entire frame* produces an easier solution, Fig. 6–34*e*. The force equations of equilibrium are the same as those above. Note that moment equilibrium will be satisfied, regardless of the radius of the pulley.